

```
# Python
print("Hello, World!")
```

```
x = 10
y = 20
```

```
def add(a, b):
    return a + b
```

```
result = add(x, y)
print("Result:", result)
```

```
for i in range(5):
    print(i)
```

CODE.
LEARN.
BUILD.
INNOVATE.



EASY SYNTAX



POWERFUL
LANGUAGE



BUILT-IN
FUNCTIONS



DATA ANALYSIS
& VISUALIZATION



AUTOMATION
& SCRIPTING



AI & MACHINE
LEARNING



WEB
DEVELOPMENT

```
Python IDE (VS Code) showing a Python program (main.py) with the following code:
```

```
1 # Python Program
2 def greet(name):
3     message = f"Hello, {name}!"
4     return message
5
6 if __name__ == "__main__":
7     name = input("Enter your name: ")
8     print(greet(name))
9
10 # Simple Addition
11 def add(a, b):
12     return a + b
13
14 result = add(10, 20)
15 print("Result:", result)
```

Terminal output:

```
Enter your name: Python Developer
Hello, Python Developer!
Result: 30
> |
```



{
Think
Code
Execute
Repeat
}



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✓ Python Kya Hai?

Python ek **super easy aur powerful** programming language hai. Ise 1991 mein Guido van Rossum naam ke bande ne banaya tha. Yeh high-level language hai, matlab iska code English jaisa hota hai – padhne aur likhne mein bahut simple. C++ ya Java jaisa nahi, jisme bahut rules aur semicolons hote hain. Python mein code clean aur short hota hai.

Python free hai, open-source hai aur duniya bhar mein sabse popular languages mein se ek hai (2026 mein bhi top pe hai).

Python Se Kya-Kya Kar Sakte Ho? (Real Uses)

Python se bahut kuch ban sakta hai bhai. Yeh almost har field mein kaam aata hai:

- **Websites aur Web Apps:** Django aur Flask use karke full websites banao (jaise Instagram, YouTube ka backend Python pe chalta hai).
- **Data Science aur Machine Learning (AI):** Pandas, NumPy, TensorFlow, Scikit-learn se data analyze karo, charts banao, AI models train karo (jaise chatbots, image recognition).
- **Automation (Roz ka kaam):** Files ko automatically organize karo, WhatsApp/Email bhejo, Excel reports banao, ya boring tasks ko 1-click mein kar do.
- **Games:** Pygame library se 2D games banao (Snake, Pong jaise).
- **Android/iOS Apps:** Kivy ya BeeWare se mobile apps.
- **Scientific Research:** Scientists aur engineers calculations, simulations, physics ke kaam karte hain.
- **Cyber Security:** Hacking tools, password crackers (ethical way mein), network scanners.
- **Blockchain, Crypto, IoT:** Bitcoin jaise tools, smart contracts, ya ghar ke smart devices control karo.
- **Web Scraping:** Kisi website se data nikaalo (jaise prices compare karna).
- Aur bahut kuch! Google, Netflix, NASA, Spotify sab Python use karte hain.

Short mein: **Agar kuch bhi computer se karwana hai toh Python se ho jaayega!**

Pehla Program – "Hello World" (Sabse Important Step)

Ab shuru karte hain asli kaam se. Har programmer ka pehla program yahi hota hai.

Code yeh hai:

Python

```
print("Hello, World!")
```

Output kya aayega? Hello, World!

(Yeh maine abhi test kiya hai – bilkul sahi chal raha hai!)

Kaise Run Kare Step-by-Step:

1. Apne laptop pe **Python install** kar lo (python.org se latest version download karo – 3.12 ya usse upar).
2. Ek naya file banao, naam rakh do hello.py.
3. Upar wala code copy-paste kar do.
4. Command Prompt (Windows) ya Terminal (Mac/Linux) kholo.

5. File wale folder mein jaao aur type karo: python hello.py Enter dabao.

✓ VARIABLES

1. Variable Kya Hai?

Variable = Memory ka naam

Computer ki memory mein bahut saari jagah hoti hai (jaise ghar mein almirah, fridge, drawer). Variable banana matlab us jagah ko ek **naam** de dena aur usme kuch value rakh dena.

Real-life Analogy (Samajhne ke liye):

- Maan lo tumhare fridge mein ek dabba hai.
- Us dabbe ka naam rakh diya **"Fruits"**.
- Aaj usme apple daal diya.
- Kal apple nikaal ke mango daal sakte ho.

Yahin **"Fruits"** = Variable name Apple / Mango = Value jo andar store hai.

Python mein bhi yahi hota hai – variable naam se hum memory mein value store karte hain.

2. Variable Kaise Banate Hain? (Syntax)

Python

```
variable_name = value
```

- Left side → Variable ka naam
- Right side → Value jo store karna hai
- = ko **assignment operator** kehte hain (value assign karna)

Live Examples:

Python

```
name = "Pawan"           # string (text)
age = 25                  # integer (poora number)
height = 5.8              # float (decimal wala number)
is_student = True         # boolean (True/False)
city = None                # None matlab abhi koi value nahi hai
```

3. Variable Ke Rules (Bahut Important!)

Python ke rules follow karne padte hain, warna error aayega:

1. Sirf ye characters use kar sakte ho:

- a. Letters (a-z, A-Z)
- b. Numbers (0-9)
- c. Underscore (_)

2. **Pehla character number nahi ho sakta** ✓ age1 = 25 ✗ 1age = 25

3. **Space allowed nahi** ✓ my_name = "Pawan" ✗ my name = "Pawan"

4. **Special symbols nahi** (@, #, \$, %, etc.)

5. **Python ke keywords nahi use kar sakte** (jaise if, else, True, False, None – ye reserved hain)

6. **Case sensitive hai** Age aur age dono alag-alag variables hain!

Best Practice (Professional tarika):

- Meaningful naam rakho → total_price, user_age, student_name
- Snake case use karo → my_name_is (common style Python mein)

4. Python Variables Dynamic Hain (Sabse Powerful Feature)

Ek baar variable bana diya toh baad mein usme kisi bhi type ki value daal sakte ho:

```
Python
x = 10          # abhi integer
x = "Hello"     # ab string ho gaya
x = 45.67       # ab float ho gaya
x = True        # ab boolean
```

Yeh Python ki khasiyat hai (dynamic typing).

5. Variables Ko Use Kaise Karte Hain?

Python

```
name = "Pawan"
age = 25
print(name)          # Output: Pawan
print("Meri umar hai", age, "saal")

# Calculation mein bhi use kar sakte ho
future_age = age + 5
print("5 saal baad umar hogi:", future_age)
```

6. Common Mistakes Beginners Karte Hain

- Variable naam mein space daal dena
- Pehle number likh dena
- Quotes bhool jaana string mein (name = Pawan galat hai)
- = ki jagah == likh dena (== comparison ke liye hota hai)

✓ DATA TYPE

1. Data Type Kya Hai?

Data Type = Value ka “jaati” ya “prakaar”

Jaise insaan ke liye alag-alag jaati hoti hai (naam, umar, salary), waise hi Python mein har cheez ka alag data type hota hai. Python khud automatically type detect kar leta hai.

Sabse important 5 Basic Data Types

Data Type	Full Name	Kya Store Karta Hai?	Example	type() ka Output
int	Integer	Poora number (decimal nahi)	25, -10, 0, 100000	<class 'int'>
float	Floating Point	Decimal wala number	25.99, 3.14, -7.5	<class 'float'>
str	String	Text / characters	"Pawan", 'Hello', "25"	<class 'str'>
bool	Boolean	Sirf 2 values: True ya False	True, False	<class 'bool'>

NoneType	None	Kuch nahi (empty)	None	<class 'NoneType'>
----------	------	-------------------	------	--------------------

2. Har Type Ko Detail Mein Samjho

1. int (Integer)

- Koi bhi poora number (positive, negative ya zero)
- Decimal nahi hota

Python

```
age = 25
marks = -5
population = 1400000000
print(type(age))      # <class 'int'>
```

2. float

- Decimal point wala number
- Scientific calculation, money, temperature ke liye best

Python

```
height = 5.8
price = 99.99
temp = -4.5
print(type(height))   # <class 'float'>
```

3. str (String)

- Text, naam, sentence, number ko bhi text bana sakte ho
- Hamesha **single** (' '), **double** (" ") ya **triple** (""" """) quotes mein likho

Python

```
name = "Pawan Kumar"
college = 'Government College'
address = """Haryana, India"""
print(type(name))      # <class 'str'>

# Number ko string bana diya
roll_no = "25"          # yeh string hai, calculation nahi ho sakti
```

4. bool (Boolean)

- Sirf True ya False (capital T aur F hona zaroori)
- Condition check karne ke liye bahut use hota hai

Python

```
is_adult = True
has_license = False
print(type(is_adult))  # <class 'bool'>
```

5. None

- Matlab “kuch nahi” ya “abhi value nahi di”

Python

```
result = None
print(type(result))    # <class 'NoneType'>
```

3. Type Check Karne Ka Sabse Best Tarika

Python

```
x = 25
y = 25.5
z = "25"

print(type(x))    # int
print(type(y))    # float
print(type(z))    # str
```

4. Type Conversion (Type Casting) – Bahut Important

Kabhi kabhi ek type ko dusre type mein badalna padta hai:

Python

```
# String ko number mein
age_str = "25"
age = int(age_str)    # ab age int ban gaya

# Number ko string mein
marks = 95
marks_str = str(marks)

# Float ko int mein (decimal cut ho jaata hai)
pi = 3.14
pi_int = int(pi)    # 3

print(type(age))    # int
```

Note: Agar string mein number nahi hai toh error aayega (int("Pawan") galat hai).

5. Common Mistakes Beginners Karte Hain

- Number ko quotes mein likh dena → woh string ban jaata hai a = "10" + "20" → "1020" (calculation nahi hogi)
- True ko true likh dena (small t) → error
- None ko none likh dena → error

✓ PRACTICE SET

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1-5: Basic Variables Creation

Q1. Ek variable name banao aur usme apna naam (string) store karo. Print karo. **Code:**

Python

```
name = "Pawan"
print(name)
```

Output:

text

Pawan

Q2. Ek variable age banao aur usme 25 store karo (int). Print karo. **Code:**

Python

```
age = 25
print(age)
```

Output:

text

25

Q3. Ek variable height banao aur usme 5.8 store karo (float). Print karo. **Code:**

Python

```
height = 5.8
print(height)
```

Output:

text

5.8

Q4. Ek variable is_student banao aur usme True store karo (bool). Print karo. **Code:**

Python

```
is_student = True
print(is_student)
```

Output:

text

True

Q5. Ek variable result banao aur usme None store karo. Print karo. **Code:**

Python

```
result = None
print(result)
```

Output:

text

None

6-10: Data Types aur type() Function

EASY SYNTAX

Q6. Upar wale sab variables banao aur unka type() print karo. **Code:**

Python

```
name = "Pawan"
age = 25
height = 5.8
is_student = True
result = None

print(type(name))
print(type(age))
print(type(height))
print(type(is_student))
print(type(result))
```

Output:

```
text
<class 'str'>
<class 'int'>
<class 'float'>
<class 'bool'>
<class 'NoneType'>
```

Q7. Ek string variable roll banao jisme "101" store ho. Uska type print karo. **Code:**

Python

```
roll = "101"
print(roll)
print(type(roll))
```

Output:

```
text
101
<class 'str'>
```

Q8. Ek variable x banao, usme 100 store karo, phir usi variable mein "Hello" store kar do. Type check karo. **Code:**

Python

```
x = 100
print(type(x))
x = "Hello"
print(type(x))
```

Output:

```
text
```



```
<class 'int'>
<class 'str'>
```

Q9. User se naam input lo aur uska type print karo. **Code:**

Python

```
name = input("Apna naam batao: ")
print(name)
print(type(name))
```

Output (example input "Pawan" diya toh):

text

Pawan

```
<class 'str'>
```

Q10. User se age input lo (string aayega), phir int() se convert karke type print karo. **Code:**

Python

```
age = input("Umar batao: ")
age = int(age)
print(age)
print(type(age))
```

Output (example input 25 diya toh):

text

25

```
<class 'int'>
```

11-15: Type Conversion (Casting)

Q11. String "456" ko integer mein convert karo aur print karo. **Code:**

Python

```
num_str = "456"
num = int(num_str)
print(num)
print(type(num))
```

Output:

text

456

```
<class 'int'>
```

Q12. Integer 99 ko string mein convert karo aur print karo. **Code:**

Python

```
marks = 99
marks_str = str(marks)
print(marks_str)
print(type(marks_str))
```

Output:

text

```
99
<class 'str'>
```

Q13. Float 3.14 ko integer mein convert karo (decimal cut ho jaayega). **Code:**

Python

```
pi = 3.14
pi_int = int(pi)
print(pi_int)
```

Output:

text

```
3
```

Q14. User se float number lo aur usko integer mein convert karke print karo. **Code:**

Python

```
num = input("Koi decimal number daalo: ")
num = float(num)
print("Float value:", num)
print("Integer value:", int(num))
```

Output (example 45.67):

text

```
Float value: 45.67
```

```
Integer value: 45
```

Q15. Boolean True ko string mein convert karke print karo. **Code:**

Python

```
status = True
status_str = str(status)
print(status_str)
print(type(status_str))
```

Output:

text

```
True
```

```
<class 'str'>
```

16-20: Mixed Practical (Variables + Types + Small Logic)

Q16. Teen variables banao: a=10, b=20.5, c="30". Sabka sum (type conversion ke baad) print karo. **Code:**

Python

```
a = 10
b = 20.5
c = "30"
total = a + b + int(c)
print("Total =", total)
```

Output:

text

```
Total = 60.5
```

Q17. Variable price = 99.99 banao. Isko integer aur string dono mein convert karke print karo. **Code:**

Python

```
price = 99.99
print(int(price))
print(str(price))
```

Output:

text

99

99.99

Q18. User se naam aur umar lo. Ek sentence mein print karo. **Code:**

Python

```
name = input("Naam: ")
age = int(input("Umar: "))
print("Hello", name, "aapki umar", age, "saal hai.")
```

Output (example):

text

Hello Pawan aapki umar 25 saal hai.

Q19. Ek variable x = 0 banao. Phir usme 5 add karo aur type check karo. **Code:**

Python

```
x = 0
x = x + 5
print(x)
print(type(x))
```

Output:

text

5

<class 'int'>

Q20. Ek program banao jisme 4 variables ho (naam, age, height, is_adult). Sab print karo aur type bhi dikhao. **Code:**

Python

```
name = "Pawan"
age = 25
height = 5.8
is_adult = True

print("Naam:", name, "Type:", type(name))
print("Umar:", age, "Type:", type(age))
print("Height:", height, "Type:", type(height))
print("Adult hai?", is_adult, "Type:", type(is_adult))
```


Output:

text

```
Naam: Pawan Type: <class 'str'>
Umar: 25 Type: <class 'int'>
Height: 5.8 Type: <class 'float'>
Adult hai? True Type: <class 'bool'>
```



STRINGS

1. String Kya Hai?

String = Text / Characters ka sequence Jaise “Pawan”, “Hello World!”, “12345” (number bhi string ban sakta hai agar quotes mein ho).

- String hamesha **quotes** mein likhi jaati hai.
- Python mein string **immutable** hoti hai (ek baar ban gayi toh change nahi kar sakte, naya string banana padta hai).

2. String Kaise Banate Hain? (3 Tarike)

Python

```
# 1. Double quotes
name = "Pawan Kumar"

# 2. Single quotes
city = 'Haryana'

# 3. Triple quotes (multi-line string ke liye best)
address = """House No. 123,
Sector 14,
Haryana, India"""

print(name)
print(city)
print(address)
```

Output:

text

```
Pawan Kumar
Haryana
House No. 123,
Sector 14,
Haryana, India
```

3. String Ke Important Properties

- Ordered → har character ka index hota hai (0 se shuru)

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- Immutable → change nahi kar sakte
- Can contain letters, numbers, symbols, spaces sab kuch

4. Basic Operations on Strings

Operation	Kya Karta Hai	Example	Output
+ (Concat)	Do strings jodta hai	"Hello" + " World"	"Hello World"
* (Repeat)	String ko kitni baar repeat	"Hi" * 3	"HiHiHi"
len()	Kitne characters hain	len("Pawan")	5

5. Indexing & Slicing (Sabse Important)

Python

```
s = "Python"

# Indexing (0 se shuru hota hai)
print(s[0])    # P
print(s[1])    # y
print(s[-1])   # n (last character)

# Slicing [start : end : step]
print(s[0:4])  # Pyth
print(s[2:])   # thon
print(s[:4])   # Pyth
print(s[::-1]) # nohtyP (reverse)
```

6. Important String Methods (Built-in Functions)

Yeh methods original string ko change nahi karte, nayi string return karte hain:

Method	Kaam	Example
upper()	Sab capital letters	"pawan".upper() → "PAWAN"
lower()	Sab small letters	"PAWAN".lower() → "pawan"
strip()	Extra spaces hata deta hai	" Hello ".strip() → "Hello"
replace(old, new)	Ek word ko dusre se replace	"Hello Pawan".replace("Pawan", "Rahul")
split()	String ko list mein tod deta hai	"Pawan Kumar".split() → ["Pawan", "Kumar"]
join()	List ko string mein jodta hai	"-".join(["Pawan", "Kumar"]) → "Pawan-Kumar"
find() / index()	Substring dhundta hai	"Python".find("th") → 2
count()	Kitni baar substring hai	"banana".count("a") → 3
startswith()	Kisi word se shuru hoti hai?	"Hello".startswith("He") → True
endswith()	Kisi word pe khatam hoti hai?	"Python".endswith("on") → True

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7. String Formatting (Modern Way – f-strings)

Python

```
name = "Pawan"
age = 25
city = "Haryana"

# Old way
print("Mera naam {} hai, umar {} hai aur main {} se hoon.".format(name, age, city))

# Best & Modern way (f-string)
print(f"Mera naam {name} hai, umar {age} hai aur main {city} se hoon.")
```

Output:

text

Mera naam Pawan hai, umar 25 hai aur main Haryana se hoon.

8. Escape Sequences (Special Characters)

- `\n` → new line
- `\t` → tab space
- `\\` → backslash
- `\"` → double quote andar likhne ke liye

Example:

Python

```
print("Hello\nWorld")    # Hello
                        # World
```

✓ PRACTICE SET

Q1. Ek string `s = "Python Programming"` banao aur uski length print karo. **Code:**

Python

```
s = "Python Programming"
print(len(s))
```

Output:

text

18

Q2. String `"hello world"` ko upper case mein convert karo. **Code:**

Python

```
print("hello world".upper())
```

Output:

text

HELLO WORLD

Q3. String `" Pawan "` ke dono taraf ke spaces hatao. **Code:**

Python

```
print(" Pawan ".strip())
```

Output:

text

Pawan

Q4. String "I love Python" mein "Python" ko "Java" se replace karo. **Code:**

Python

```
print("I love Python".replace("Python", "Java"))
```

Output:

text

I love Java

Q5. String "Pawan Kumar Sharma" ko words mein split karo. **Code:**

Python

```
print("Pawan Kumar Sharma".split())
```

Output:

text

['Pawan', 'Kumar', 'Sharma']

Q6. User se naam lo aur usko title case mein print karo (title() method). **Code:**

Python

```
name = input("Naam: ")  
print(name.title())
```

Output (agar "pawan kumar" daala):

text

Pawan Kumar

Q7. String "banana" mein 'a' kitni baar hai? **Code:**

Python

```
print("banana".count("a"))
```

Output:

text

3

Q8. String "Python" ko reverse karo (slicing se). **Code:**

Python

```
print("Python"[::-1])
```

Output:

text

nohtyP

Q9. Do strings "Hello" aur "World" ko space ke saath jodo. **Code:**

Python

```
print("Hello" + " " + "World")
```

Output:

text

```
Hello World
```

Q10. String "12345" ko integer mein convert karo aur type check karo. **Code:**

Python

```
print(int("12345"))  
print(type(int("12345")))
```

Output:

text

```
12345  
<class 'int'>
```

Q11. f-string use karke: naam, age aur city print karo. **Code:**

Python

```
naam = "Pawan"  
age = 25  
city = "Haryana"  
print(f"Mera naam {naam} hai, umar {age} saal hai aur main {city} se hoon.")
```

Output:

text

```
Mera naam Pawan hai, umar 25 saal hai aur main Haryana se hoon.
```

Q12. String "hello@world@python" ko @ se split karo. **Code:**

Python

```
print("hello@world@python".split("@"))
```

Output:

text

```
['hello', 'world', 'python']
```

Q13. String " Python " ke baad aur pehle spaces hatao aur length print karo. **Code:**

Python

```
s = " Python "  
clean = s.strip()  
print(clean)  
print("Length =", len(clean))
```

Output:

text

```
Python  
Length = 6
```

Q14. User se sentence lo aur usme "a" kitni baar hai count karo. **Code:**

Python

```
sentence = input("Ek sentence likho: ")  
print("'a' kitni baar hai:", sentence.lower().count("a"))
```

Output (example input: "I am a Python programmer"):

text

```
'a' kitni baar hai: 3
```

Q15. String "I am learning Python" mein "learning" shuru hota hai ya nahi check karo. **Code:**

Python `b):`

```
s = "I am learning Python"
print(s[5:].startswith("learning"))
```

Output:

text

```
True
```

Q16. Ek list ["Pawan", "Rahul", "Aman"] ko comma se join karke string banao. **Code:**

Python

```
names = ["Pawan", "Rahul", "Aman"]
print(", ".join(names))
```

Output:

text

```
Pawan, Rahul, Aman
```

Q17. String "P@wan#123" mein sirf alphanumeric characters rakho (isalnum()). **Code:**

Python

```
s = "P@wan#123"
clean = ''.join(c for c in s if c.isalnum())
print(clean)
```

Output:

text

```
Pwan123
```

Q18. String ko 3 baar repeat karo using *. **Code:**

Python

```
print("Hello! " * 3)
```

Output:

text

```
Hello! Hello! Hello!
```

Q19. Multi-line string banao aur usme \n use karke lines break karo. **Code:**

Python

```
print("Line 1\nLine 2\nLine 3\nYeh multi-line string hai")
```

Output:

text

```
Line 1
```

```
Line 2
```

```
Line 3
```

```
Yeh multi-line string hai
```

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Q20. User se full name lo aur uske first aur last name ko alag-alag print karo. **Code:**

Python

```
full = input("Apna full naam batao: ")
parts = full.split()
print("First Name :", parts[0])
print("Last Name  :", parts[-1])
```

Output (agar "Pawan Kumar Sharma" daala):

text

```
First Name : Pawan
Last Name  : Sharma
```

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✓ CONDITIONAL STATEMENTS

1. Conditional Statements Kya Hai?

- Computer ko bolna: "Agar yeh condition true hai toh yeh kar, warna woh kar."
- Real life example:
 - Agar barish ho rahi hai toh chhatri leke jaao.
 - Warna bina chhatri ke jaao.

Python mein 3 main keywords hote hain:

- if → agar
- elif → aur agar (else if)
- else → warna

2. Basic Syntax (Sabse Important)

Python

```
if condition:
    # yeh code tabhi chalega jab condition True ho
    print("Condition sahi hai")
elif condition2:
    # agar pehli condition galat ho aur yeh sahi ho
    print("Doosri condition sahi hai")
else:
    # agar upar ki dono conditions galat ho
    print("Koi bhi condition sahi nahi hai")
```

Note:

- : colon lagana zaroori hai
- Indentation (4 spaces) bahut important hai – galat indentation se error aayega

3. Comparison Operators (Condition banane ke liye)

Operator	Matlab	Example
==	barabar hai	age == 18

!=	barabar nahi hai	age != 18
>	bada hai	age > 18
<	chhota hai	age < 18
>=	bada ya barabar	age >= 18
<=	chhota ya barabar	age <= 18

4. Logical Operators (Multiple conditions)

- and → dono conditions sahi honi chahiye
- or → ek bhi condition sahi ho toh chalega
- not → ulat kar deta hai

5. Full Practical Examples

Example 1: Simple if-else

Python

```
age = 20

if age >= 18:
    print("Aap adult ho, voting kar sakte ho")
else:
    print("Aap minor ho")
```

Output: Aap adult ho, voting kar sakte ho

Example 2: if-elif-else

Python

```
marks = 85

if marks >= 90:
    print("Grade A")
elif marks >= 75:
    print("Grade B")
elif marks >= 60:
    print("Grade C")
else:
    print("Fail")
```

Output: Grade B

Example 3: Multiple conditions with and/or

Python

```
age = 25
has_license = True

if age >= 18 and has_license:
```



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```
print("Aap car chala sakte ho")
else:
    print("Nahi chala sakte")
```

Example 4: Nested if (if ke andar if)

Python

```
num = 0

if num > 0:
    print("Positive")
elif num < 0:
    print("Negative")
else:
    print("Zero")
```

6. Real Useful Programs (Copy-paste karke try karo)

Python

```
# Program 1: Even-Odd Checker
num = int(input("Koi number daalo: "))
if num % 2 == 0:
    print("Even number hai")
else:
    print("Odd number hai")
```

Python

```
# Program 2: Biggest number among 3
a = int(input("Pehla number: "))
b = int(input("Doosra number: "))
c = int(input("Teesra number: "))

if a > b and a > c:
    print(a, "sabse bada hai")
elif b > c:
    print(b, "sabse bada hai")
else:
    print(c, "sabse bada hai")
```



PRACTICE SET

Q1. User se age input lo. Agar 18 ya usse zyada hai toh "Adult" print karo, warna "Minor". **Code:**

Python

```
age = int(input("Umar batao: "))
if age >= 18:
    print("Adult")
```



```
else:  
    print("Minor")
```

Output (age 20 daalo toh): Adult

Q2. User se number lo aur even ya odd batao. **Code:** (upar wala even-odd program)

Q3. User se marks lo aur grade batao (90=A, 75=B, 60=C, else Fail). **Code:** (upar wala Grade program)

Q4. User se 3 numbers lo aur sabse bada number print karo. **Code:** (upar wala biggest number program)

Q5. User se temperature lo. Agar >30 hai toh "Garmi hai", 20-30 ke beech "Normal", <20 toh "Thand hai".

Code:

Python

```
temp = float(input("Temperature batao: "))  
if temp > 30:  
    print("Garmi hai")  
elif temp >= 20:  
    print("Normal hai")  
else:  
    print("Thand hai")
```

Q6. User se number lo. Agar 0 se bada hai toh "Positive", chhota hai toh "Negative", 0 hai toh "Zero"

Code:

Python

```
num = int(input("Koi number daalo: "))  
  
if num > 0:  
    print("Positive")  
elif num < 0:  
    print("Negative")  
else:  
    print("Zero")
```

Output (example inputs):

- Input: 5 → Positive
- Input: -3 → Negative
- Input: 0 → Zero

Q7. User se 2 numbers lo aur check karo dono barabar hain ya nahi. **Code:**

Python

```
a = int(input("Pehla number: "))  
b = int(input("Doosra number: "))  
  
if a == b:  
    print("Dono numbers barabar hain")
```

```
else:
    print("Dono numbers alag hain")
```

Output (example):

- Input: 10 and 10 → Dono numbers barabar hain
- Input: 10 and 20 → Dono numbers alag hain

Q8. User se password lo. Agar "python123" hai toh "Login successful", warna "Wrong password". **Code:**

Python

```
password = input("Password daalo: ")

if password == "python123":
    print("Login successful")
else:
    print("Wrong password")
```

Output:

- Correct password daalo → Login successful
- Galat password daalo → Wrong password

Q9. User se age aur gender lo. Agar male aur age $\geq$ 21 hai toh "Shaadi ke layak", female aur age $\geq$ 18 hai toh "Shaadi ke layak", warna "Nahi". **Code:**

Python

```
age = int(input("Umar batao: "))
gender = input("Gender (male/female): ").lower()

if gender == "male" and age >= 21:
    print("Shaadi ke layak")
elif gender == "female" and age >= 18:
    print("Shaadi ke layak")
else:
    print("Nahi")
```

Output (example):

- age=22, male → Shaadi ke layak
- age=17, female → Nahi

Q10. User se number lo aur check karo 5 se divisible hai ya nahi. **Code:**

Python

```
num = int(input("Koi number daalo: "))

if num % 5 == 0:
    print("Number 5 se divisible hai")
```

```
else:  
    print("Number 5 se divisible nahi hai")
```

Output:

- Input: 25 → Number 5 se divisible hai
- Input: 27 → Number 5 se divisible nahi hai

Q11. User se 3 numbers lo aur unka average nikaalo. Agar average >60 hai toh "Pass", warna "Fail".

Code:

Python

```
a = int(input("Pehla number: "))  
b = int(input("Doosra number: "))  
c = int(input("Teesra number: "))
```

```
average = (a + b + c) / 3
```

```
if average > 60:  
    print("Pass")  
else:  
    print("Fail")
```

Output (example):

- Numbers: 70 80 90 → Pass
- Numbers: 40 50 55 → Fail

Q12. User se character lo aur check karo vowel hai ya consonant. **Code:**

Python

```
ch = input("Ek character daalo: ").lower()  
  
if ch in "aeiou":  
    print("Vowel hai")  
elif ch.isalpha():  
    print("Consonant hai")  
else:  
    print("Character letter nahi hai")
```

Output:

- Input: a → Vowel hai
- Input: b → Consonant hai

Q13. User se salary lo. Agar >50000 hai toh 10% tax, 30000-50000 ke beech 5% tax, warna no tax. **Code:**

Python


```

salary = float(input("Salary batao: "))

if salary > 50000:
    tax = salary * 0.10
    print("Tax =", tax)
elif salary >= 30000:
    tax = salary * 0.05
    print("Tax =", tax)
else:
    print("No tax")

```

Output (example):

- Salary 60000 → Tax = 6000.0
- Salary 25000 → No tax

Q14. User se year lo aur check karo leap year hai ya nahi. **Code:**

Python

```

year = int(input("Year batao: "))

if (year % 4 == 0 and year % 100 != 0) or (year % 400 == 0):
    print("Leap year hai")
else:
    print("Leap year nahi hai")

```

Output:

- Year 2024 → Leap year hai
- Year 2023 → Leap year nahi hai

Q15. User se 2 numbers aur ek operator (+, -, *, /) lo aur calculation karo. **Code:**

Python

```

num1 = float(input("Pehla number: "))
num2 = float(input("Doosra number: "))
op = input("Operator (+, -, *, /): ")

if op == "+":
    print(num1 + num2)
elif op == "-":
    print(num1 - num2)
elif op == "*":
    print(num1 * num2)
elif op == "/":
    if num2 != 0:

```

```

        print(num1 / num2)
    else:
        print("Division by zero nahi ho sakti")
else:
    print("Galat operator daala")

```

Output (example):

- 10, 5, * → 50.0



LIST

1. List Kya Hai?

List = Ek se zyada values ko ek saath store karne ka tarika

- Ordered hota hai (index se access kar sakte ho)
- Change kar sakte ho (mutable)
- Different types ke data rakh sakte ho (int, string, float, bool, even dusri list bhi)

Real life example:

- Tumhari shopping list → ["Doodh", "Roti", "Aalu", "Pyaz"]
- Yeh ek list hai jisme 4 cheezein hain.

2. List Kaise Banate Hain?

Python

```

# Empty list
empty = []

# Normal list
numbers = [10, 20, 30, 40]
fruits = ["Apple", "Banana", "Mango", "Orange"]
mixed = [25, "Pawan", 5.8, True, None]

print(fruits)
print(type(fruits))  # <class 'list'>

```

3. Important Properties

- Indexing shuru hota hai 0 se
- Negative indexing bhi hota hai (-1 = last element)
- Duplicate values allowed hain
- List mutable hai (change kar sakte ho)

4. Indexing & Slicing (Strings jaisa hi)

Python

```
fruits = ["Apple", "Banana", "Mango", "Orange", "Grapes"]
```

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```

print(fruits[0])      # Apple
print(fruits[2])      # Mango
print(fruits[-1])     # Grapes (Last)

# Slicing
print(fruits[1:4])    # ['Banana', 'Mango', 'Orange']
print(fruits[:3])     # ['Apple', 'Banana', 'Mango']
print(fruits[2:])     # ['Mango', 'Orange', 'Grapes']
print(fruits[::-1])   # reverse list

```

5. List Ke Sabse Important Methods

Method	Kaam kya karta hai	Example
append()	End mein ek item add karta hai	fruits.append("Kiwi")
insert()	Kisi index pe item add karta hai	fruits.insert(1, "Cherry")
remove()	Pehli occurrence delete karta hai	fruits.remove("Banana")
pop()	Last item delete aur return karta hai	fruits.pop()
pop(index)	Specific index delete	fruits.pop(2)
clear()	Saari list khali kar deta hai	fruits.clear()
sort()	Ascending order mein sort	numbers.sort()
reverse()	List ko ulta kar deta hai	fruits.reverse()
index()	Item ka index batata hai	fruits.index("Mango")
count()	Item kitni baar hai	fruits.count("Apple")
copy()	List ka copy bana deta hai	new = fruits.copy()

6. List Operations

Python

```

# Concatenation
list1 = [1, 2, 3]
list2 = [4, 5]
print(list1 + list2)      # [1, 2, 3, 4, 5]

# Repeat
print([10] * 4)           # [10, 10, 10, 10]

# Length
print(len(fruits))        # kitne items hain

```

7. Looping through List (for loop)

Python

```

fruits = ["Apple", "Banana", "Mango"]

```



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```
for fruit in fruits:  
    print(fruit)
```



PRACTICE SET

Q1. Ek list banao jisme 5 fruits ho aur print karo. **Code:**

Python

```
fruits = ["Apple", "Banana", "Mango", "Orange", "Grapes"]  
print(fruits)
```

Output:

text

```
['Apple', 'Banana', 'Mango', 'Orange', 'Grapes']
```

Q2. List ke last element ko print karo (negative indexing se). **Code:**

Python

```
fruits = ["Apple", "Banana", "Mango", "Orange", "Grapes"]  
print(fruits[-1])
```

Output:

text

```
Grapes
```

Q3. List ke 2nd aur 3rd element ko slicing se print karo. **Code:**

Python

```
fruits = ["Apple", "Banana", "Mango", "Orange", "Grapes"]  
print(fruits[1:3])
```

Output:

text

```
['Banana', 'Mango']
```

Q4. List mein ek naya fruit "Kiwi" end mein add karo (append). **Code:**

Python

```
fruits = ["Apple", "Banana", "Mango", "Orange", "Grapes"]  
fruits.append("Kiwi")  
print(fruits)
```

Output:

text

```
['Apple', 'Banana', 'Mango', 'Orange', 'Grapes', 'Kiwi']
```

Q5. List ke 2nd position pe "Cherry" insert karo. **Code:**

Python

```
fruits = ["Apple", "Banana", "Mango", "Orange", "Grapes"]  
fruits.insert(1, "Cherry")  
print(fruits)
```

Output:

text

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```
['Apple', 'Cherry', 'Banana', 'Mango', 'Orange', 'Grapes']
```

Q6. User se 5 numbers input lo aur unko list mein store karo. **Code:**

Python

```
nums = []
for i in range(5):
    x = int(input(f"Number {i+1} daalo: "))
    nums.append(x)
print(nums)
```

Output (example input 10 20 30 40 50):

text

```
[10, 20, 30, 40, 50]
```

Q7. List se "Banana" ko remove karo. **Code:**

Python

```
fruits = ["Apple", "Banana", "Mango", "Orange", "Grapes"]
fruits.remove("Banana")
print(fruits)
```

Output:

text

```
['Apple', 'Mango', 'Orange', 'Grapes']
```

Q8. List se last item pop karke print karo. **Code:**

Python

```
fruits = ["Apple", "Banana", "Mango", "Orange", "Grapes"]
print("Popped item:", fruits.pop())
print("Remaining list:", fruits)
```

Output:

text

```
Popped item: Grapes
```

```
Remaining list: ['Apple', 'Banana', 'Mango', 'Orange']
```

Q9. List ko sort karo (numbers wali list). **Code:**

Python

```
numbers = [50, 10, 80, 30, 20]
numbers.sort()
print(numbers)
```

Output:

text

```
[10, 20, 30, 50, 80]
```

Q10. List ko reverse karo. **Code:**

Python

```
fruits = ["Apple", "Banana", "Mango", "Orange", "Grapes"]
fruits.reverse()
print(fruits)
```

Output:

```
['Grapes', 'Orange', 'Mango', 'Banana', 'Apple']
```

Q11. List mein kisi item ka index dhundo. Code:

Python

```
fruits = ["Apple", "Banana", "Mango", "Orange", "Grapes"]
print(fruits.index("Mango"))
```

Output:

text

2

Q12. List mein kisi item ki count nikalo. Code:

Python

```
fruits = ["Apple", "Banana", "Apple", "Mango", "Apple"]
print(fruits.count("Apple"))
```

Output:

text

3

Q13. Do lists ko concatenate karke naya list banao. Code:

Python

```
list1 = [1, 2, 3]
list2 = [4, 5, 6]
new_list = list1 + list2
print(new_list)
```

Output:

text

```
[1, 2, 3, 4, 5, 6]
```

Q14. Ek list copy karke dusri list banao (copy()). Code:

Python

```
fruits = ["Apple", "Banana", "Mango"]
new_fruits = fruits.copy()
print(new_fruits)
```

Output:

text

```
['Apple', 'Banana', 'Mango']
```

Q15. List ko clear kar do. Code:

Python


```
fruits = ["Apple", "Banana", "Mango"]
fruits.clear()
print(fruits)
```

Output:

```
text
[]
```

Q16. User se 3 subjects ke marks lo aur unko list mein store karke average print karo. **Code:**

Python

```
marks = []
for i in range(3):
    m = int(input(f"Subject {i+1} marks: "))
    marks.append(m)
avg = sum(marks) / 3
print("Average =", avg)
```

Output (example 80 90 70):

```
text
Average = 80.0
```

Q17. List mein se sabse bada aur sabse chhota number print karo. **Code:**

Python

```
numbers = [45, 78, 12, 95, 23]
print("Sabse bada:", max(numbers))
print("Sabse chhota:", min(numbers))
```

Output:

```
text
Sabse bada: 95
Sabse chhota: 12
```

Q18. List ko for loop se print karo (har item alag line pe). **Code:**

Python

```
fruits = ["Apple", "Banana", "Mango", "Orange"]
for fruit in fruits:
    print(fruit)
```

Output:

```
text
Apple
Banana
Mango
Orange
```

Q19. Nested list banao → student = ["Pawan", 25, ["Math", "Science", "English"]]. **Code:**

Python

```
student = ["Pawan", 25, ["Math", "Science", "English"]]
print(student)
print("Subjects:", student[2])
```

Output:

```
text (a, b):
['Pawan', 25, ['Math', 'Science', 'English']]
Subjects: ['Math', 'Science', 'English']
```

Q20. User se full name lo, split karke list mein store karo aur first aur last name print karo. **Code:**

```
Python
full_name = input("Apna full naam batao: ")
name_list = full_name.split()
print("First Name :", name_list[0])
print("Last Name :", name_list[-1])
```

Output (example "Pawan Kumar Sharma"):

```
text
First Name : Pawan
Last Name : Sharma
```

✓ TUPLE

1. Tuple Kya Hai? (Simple Definition)

Tuple = Fixed (pakka) list

- Ordered hota hai (index se access kar sakte ho)
- **Immutable** → create karne ke baad andar kuch add, remove ya change nahi kar sakte
- Lists se faster hota hai
- Jab data ko change nahi karna ho (jaise coordinates, RGB color, fixed records) tab tuple use karte hain

Real life example:

- Aadhaar card ka data → naam, number, DOB (yeh change nahi hona chahiye)

2. Tuple Kaise Banate Hain?

Python

```
# Empty tuple
empty = ()

# Normal tuple
numbers = (10, 20, 30, 40)
fruits = ("Apple", "Banana", "Mango", "Orange")
mixed = (25, "Pawan", 5.8, True, None)

# Single element wala tuple (important!)
```

```
single = (5,)          # comma zaroori hai
wrong = (5)            # yeh integer ban jaayega, tuple nahi

print(type(fruits))    # <class 'tuple'>
```

3. Important Properties

- Parentheses () mein banate hain
- Ordered + Indexed
- **Immutable** (change nahi kar sakte)
- Duplicate values allowed
- Faster than list

4. Indexing & Slicing (List jaisa hi)

Python

```
fruits = ("Apple", "Banana", "Mango", "Orange", "Grapes")

print(fruits[0])      # Apple
print(fruits[-1])     # Grapes
print(fruits[1:4])    # ('Banana', 'Mango', 'Orange')
print(fruits[::-1])   # reverse
```

5. Tuple Ke Methods (Bahut Kam hain kyunki immutable hai)

Method	Kaam	Example
count()	Item kitni baar hai	fruits.count("Apple")
index()	Item ka index batata hai	fruits.index("Mango")
len()	Kitne elements hain	len(fruits)

Note: append, remove, pop, insert, sort, reverse jaise methods **nahi** hain.

6. Tuple Operations

Python

```
t1 = (1, 2, 3)
t2 = (4, 5)

print(t1 + t2)      # Concatenation
print(t1 * 3)       # Repeat
print(len(t1))      # Length
```

7. Tuple Unpacking (Bahut Powerful Feature)

Python

```
person = ("Pawan", 25, "Haryana")
name, age, city = person

print(name)         # Pawan
```

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```
print(age)    # 25
print(city)   # Haryana
```

8. List vs Tuple (Quick Comparison)

Feature	List	Tuple
Mutable	Yes	No
Speed	Slow	Fast
Brackets	[]	()
Use	Change karna ho	Fixed data

Ab 20 Practical Questions (Tuple pe) – Code + Output ke saath

Q1. Ek tuple banao jisme 5 fruits ho aur print karo. **Code:**

Python

```
fruits = ("Apple", "Banana", "Mango", "Orange", "Grapes")
print(fruits)
```

Output:

text

```
('Apple', 'Banana', 'Mango', 'Orange', 'Grapes')
```

Q2. Tuple ke last element ko negative indexing se print karo. **Code:**

Python

```
print(fruits[-1])
```

Output:

text

```
Grapes
```

Q3. Tuple ke 2nd aur 3rd element ko slicing se print karo. **Code:**

Python

```
print(fruits[1:3])
```

Output:

text

```
('Banana', 'Mango')
```

Q4. Single element wala tuple banao (5,) aur type check karo. **Code:**

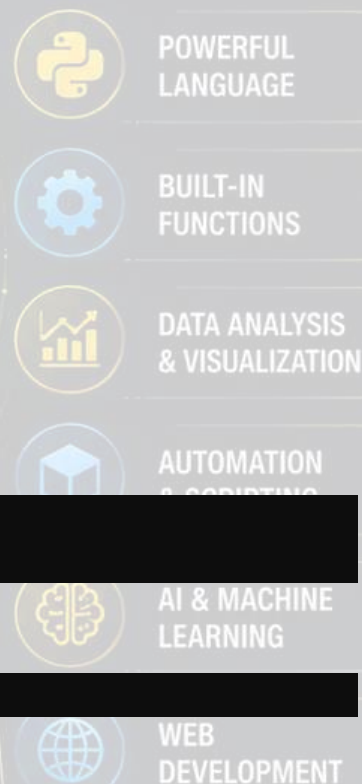
Python

```
t = (5,)
print(t)
print(type(t))
```

Output:

text

```
(5,)
<class 'tuple'>
```



Q5. Tuple mein "Mango" ka index dhundo. **Code:**

Python

```
print(fruits.index("Mango"))
```

Output:

text

2

Q6. User se 5 numbers input lo aur unko tuple mein convert karke print karo. **Code:**

Python

```
nums = []
for i in range(5):
    x = int(input(f"Number {i+1}: "))
    nums.append(x)
t = tuple(nums)
print(t)
```

Q7. Tuple mein kisi item ki count nikalo. **Code:**

Python

```
t = (1, 2, 3, 2, 4, 2, 5)
print(t.count(2))
```

Output:

text

3

Q8. Do tuples ko concatenate karke naya tuple banao. **Code:**

Python

```
t1 = (1, 2, 3)
t2 = (4, 5)
print(t1 + t2)
```

Output:

text

(1, 2, 3, 4, 5)

Q9. Tuple ko 3 baar repeat karo. **Code:**

Python

```
print(("Hi",) * 3)
```

Output:

text

('Hi', 'Hi', 'Hi')

Q10. Tuple unpacking karo (name, age, city). **Code:**

Python

```
person = ("Pawan", 25, "Haryana")
name, age, city = person
print(name, age, city)
```

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Output:

text

Pawan 25 Haryana

Q11. Tuple se list banao aur list ko sort karo. **Code:**

Python `b):`

```
t = (50, 10, 80, 30)
l = list(t)
l.sort()
print(l)
```

Q12. Tuple ki length print karo. **Code:**

Python

```
print(len(fruits))
```

Q13. Nested tuple banao. **Code:**

Python

```
student = ("Pawan", 25, ("Math", "Science", "English"))
print(student[2])
```

Q14. Tuple ko for loop se print karo. **Code:**

Python

```
for fruit in fruits:
    print(fruit)
```

Q15. Tuple ko reverse slicing se print karo. **Code:**

Python

```
print(fruits[::-1])
```

Q16. User se 3 numbers lo aur unko tuple mein store karke sum print karo.

Q17. Tuple mein se sabse bada aur sabse chhota element print karo (max() aur min()).

Q18. Do tuples compare karo (==, >, <).

Q19. Tuple ko string mein convert karo (str()).

Q20. User se full name lo, split karke tuple mein store karo aur first aur last name print karo.

✓ Dictionary

1. Dictionary Kya Hai? (Sabse Simple Definition)

Dictionary = Key-Value Pair ka Collection

- Jaise real dictionary mein **word : matlab** hota hai, waise hi Python mein **key : value** hota hai.
- Key unique hoti hai (ek hi key do baar nahi aa sakti).
- Value kuch bhi ho sakta hai (number, string, list, even dusra dictionary).
- Dictionary **mutable** hai → bana ke baad change kar sakte ho.
- Syntax: {key: value} (curly braces mein)

Real Life Example:

- Student ka record → {"name": "Pawan", "age": 25, "marks": 85}

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{
 {
 Think
 Code
 Execute
 Repeat
 }
}

2. Dictionary Kaise Banate Hain?

Python

1. Empty dictionary

```
d1 = {}
```

```
d2 = dict()
```

2. Values ke saath

```
student = {  
    "name": "Pawan",  
    "age": 25,  
    "city": "Haryana",  
    "marks": 85  
}
```

```
print(student)
```

Output:

text

```
{'name': 'Pawan', 'age': 25, 'city': 'Haryana', 'marks': 85}
```

3. Important Properties

- Keys → string, number, tuple (immutable) ho sakte hain
- Values → kuch bhi ho sakta hai
- Ordered (Python 3.7 ke baad)
- Duplicate keys allowed nahi

4. Accessing Values

Python

```
student = {"name": "Pawan", "age": 25, "marks": 85}
```

```
print(student["name"])    # Pawan
```

```
print(student["marks"])   # 85
```

Safe tarika (agar key na mile to error nahi aayega)

```
print(student.get("age"))    # 25
```

```
print(student.get("phone", "Not Found")) # Not Found
```

5. Dictionary Ke Important Methods (Table)

Method	Kaam Kya Karta Hai	Example
keys()	Saare keys dega	student.keys()
values()	Saare values dega	student.values()
items()	Saare (key, value) pairs dega	student.items()

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get(key, default)	Value deta hai, na mile to default	student.get("phone", "Nahi hai")
update()	Dusre dict ko merge kar deta hai	d1.update(d2)
pop(key)	Key delete karta hai aur value return karta	student.pop("age")
popitem()	Last pair delete karta hai	student.popitem()
clear()	Saara dictionary khali kar deta hai	student.clear()
copy()	Copy bana deta hai	new = student.copy()

6. Adding / Updating / Deleting

Python

```
student = {"name": "Pawan", "age": 25}
```

```
# Add new key
```

```
student["city"] = "Haryana"
```

```
# Update existing
```

```
student["age"] = 26
```

```
# Delete
```

```
student.pop("age")
```

```
print(student)
```

7. Looping in Dictionary

Python

```
student = {"name": "Pawan", "age": 25, "city": "Haryana"}
```

```
for key in student:
    print(key, "->", student[key])
```

```
# Best way (using items)
```

```
for k, v in student.items():
    print(f"{k} = {v}")
```

8. Nested Dictionary (Real-life use)

Python

```
student = {
    "name": "Pawan",
    "marks": {
        "math": 90,
        "science": 85,
```



```

        "english": 92
    }
}

print(student["marks"]["science"]) # 85

```

9. Dictionary Comprehension (Short & Powerful)

```

Python
# 1 se 5 tak numbers ka square
sq = {x: x*x for x in range(1, 6)}
print(sq)

```

Output:

```

text
{1: 1, 2: 4, 3: 9, 4: 16, 5: 25}

```

✓ PRACTICE SET

Q1. Ek empty dictionary banao aur usme student ka naam aur marks add karo.

```

Python
student = {}
student["name"] = "Pawan"
student["marks"] = 85
print(student)

```

Output:

```

text
{'name': 'Pawan', 'marks': 85}

```

Q2. Ek dictionary banao jisme 3 students ke naam aur unke marks ho. Print karo.

```

Python
marks = {"Pawan": 85, "Rahul": 92, "Aman": 78}
print(marks)

```

Output:

```

text
{'Pawan': 85, 'Rahul': 92, 'Aman': 78}

```

Q3. Upar wale dictionary se "Rahul" ke marks print karo.

```

Python
marks = {"Pawan": 85, "Rahul": 92, "Aman": 78}
print(marks["Rahul"])

```

Output:

```

text
92

```

Q4. Dictionary ki length (kitne pairs hain) print karo.

```

Python

```

```
marks = {"Pawan": 85, "Rahul": 92, "Aman": 78}
print(len(marks))
```

Output:

text

3

Q5. Dictionary mein naya student "Sita" add karo aur uske marks 95 daalo.

Python

```
marks = {"Pawan": 85, "Rahul": 92, "Aman": 78}
marks["Sita"] = 95
print(marks)
```

Output:

text

```
{'Pawan': 85, 'Rahul': 92, 'Aman': 78, 'Sita': 95}
```

Methods & Operations (6-10)

Q6. get() method se "Rahul" ke marks lo (agar key na mile toh default value do).

Python

```
marks = {"Pawan": 85, "Rahul": 92, "Aman": 78}
print(marks.get("Rahul"))
print(marks.get("Rohan", "Not Found"))
```

Output:

text

92

Not Found

Q7. Dictionary ke saare keys aur values alag-alag print karo.

Python

```
marks = {"Pawan": 85, "Rahul": 92, "Aman": 78}
print("Keys:", list(marks.keys()))
print("Values:", list(marks.values()))
```

Output:

text

```
Keys: ['Pawan', 'Rahul', 'Aman']
```

```
Values: [85, 92, 78]
```

Q8. items() method se saare key-value pairs print karo.

Python

```
marks = {"Pawan": 85, "Rahul": 92, "Aman": 78}
print(marks.items())
```

Output:

text

```
dict_items([('Pawan', 85), ('Rahul', 92), ('Aman', 78)])
```

Q9. "Aman" ke marks ko 88 kar do (update).

Python

```
marks = {"Pawan": 85, "Rahul": 92, "Aman": 78}
marks["Aman"] = 88
print(marks)
```

Output:

text

```
{'Pawan': 85, 'Rahul': 92, 'Aman': 88}
```

Q10. "Rahul" ko dictionary se delete kar do (pop() use karke).

Python

```
marks = {"Pawan": 85, "Rahul": 92, "Aman": 78}
marks.pop("Rahul")
print(marks)
```

Output:

text

```
{'Pawan': 85, 'Aman': 78}
```

Advanced / Real-life Questions (11-15)

Q11. User se naam aur marks input lo aur dictionary mein add karo.

Python

```
data = {}
name = input("Naam: ")
marks = int(input("Marks: "))
data[name] = marks
print(data)
```

Output (example input):

text

```
{'Pawan': 85}
```

Q12. Ek nested dictionary banao (student ka naam, age aur city).

Python

```
student = {
    "name": "Pawan",
    "age": 25,
    "city": "Haryana",
    "marks": {"math": 90, "science": 85}
}
print(student["marks"]["math"])
```

Output:

text

```
90
```

Q13. Dictionary ko loop karke saare students aur unke marks print karo.

Python

```
marks = {"Pawan": 85, "Rahul": 92, "Aman": 78}
for name, m in marks.items():
    print(f"{name} ke marks: {m}")
```

Output:

```
Pawan ke marks: 85
Rahul ke marks: 92
Aman ke marks: 78
```

Q14. Do dictionaries ko merge (combine) karo (update() use karke).

Python

```
d1 = {"Pawan": 85, "Rahul": 92}
d2 = {"Aman": 78, "Sita": 95}
d1.update(d2)
print(d1)
```

Output:

```
{'Pawan': 85, 'Rahul': 92, 'Aman': 78, 'Sita': 95}
```

Q15. Dictionary se saare keys ko list mein convert karke print karo.

Python

```
marks = {"Pawan": 85, "Rahul": 92, "Aman": 78}
key_list = list(marks.keys())
print(key_list)
```

Output:

```
text
['Pawan', 'Rahul', 'Aman']
```

✓ SET

1. Set Kya Hai? (Sabse Simple Definition)

Set = Unique cheezon ka collection (koi duplicate nahi allowed)

- Jaise real life mein "set of pens" mein ek hi pen do baar nahi hota.
- Set mein har element **unique** hota hai.
- Set **unordered** hota hai (koi fixed order nahi).
- Set **mutable** hota hai (bana ke baad change kar sakte ho).
- Syntax: {} curly braces mein likhte hain (dictionary se alag kyunki isme key-value nahi, sirf values).

Real Life Example:

- Class mein unique roll numbers ka set → {1, 5, 7, 9, 12}

2. Set Kaise Banate Hain?

Python

1. Direct curly braces se

```
s1 = {1, 2, 3, 4, 5}
```

2. set() function se (list ya string se bana sakte ho)

```
s2 = set([10, 20, 30, 10, 20]) # duplicates automatically remove ho jaayenge
```

```
s3 = set("Pawan") # har character alag
```

```
print(s1)
```

```
print(s2)
```

```
print(s3)
```

Output:

text

```
{1, 2, 3, 4, 5}
```

```
{10, 20, 30}
```

```
{'P', 'a', 'w', 'n'}
```

Note: Empty set banane ke liye set() use karo, {} se empty dictionary ban jaata hai.

3. Set Ke Important Properties

- No duplicates allowed
- Unordered (indexing nahi kar sakte → s[0] galat hai)
- Mutable (add/remove kar sakte ho)
- Fast membership testing (kisi cheez set mein hai ya nahi → bahut jaldi check)

4. Set Ke Important Methods (Table)

Method	Kaam Kya Karta Hai	Example
add()	Ek element add karta hai	s.add(6)
update()	Multiple elements add karta hai	s.update([7, 8, 9])
remove()	Element delete (agar na mile to error)	s.remove(3)
discard()	Element delete (agar na mile to error nahi)	s.discard(99)
pop()	Koi bhi ek element delete karta hai	s.pop()
clear()	Saara set khali kar deta hai	s.clear()
union()	Do sets ko jodta hai ()	
intersection()	Common elements (&)	s1 & s2 ya s1.intersection(s2)
difference()	Sirf pehle set mein jo nahi hai dusre mein	s1 - s2
symmetric_difference()	Jo dono mein common nahi hai	s1 ^ s2
issubset()	Kya ek set dusre ka subset hai?	s1.issubset(s2)

5. Practical Examples

Python

```
s = {1, 2, 3, 4, 5, 5, 3}    # duplicates remove ho gaye
print(s)                    # {1, 2, 3, 4, 5}

s.add(6)
s.update([7, 8])
print(s)

# Mathematical operations
a = {1, 2, 3, 4}
b = {3, 4, 5, 6}

print(a | b)                # union → {1,2,3,4,5,6}
print(a & b)                # intersection → {3,4}
print(a - b)                # difference → {1,2}
```

✓ Practice set

Q1. Ek set banao jisme 5 unique numbers ho.

Python

```
s = {10, 20, 30, 40, 50}
print(s)
```

Output:

text

```
{10, 20, 30, 40, 50}
```

Q2. List [1, 2, 2, 3, 4, 4, 5] ko set mein convert karke duplicates remove karo.

Python

```
lst = [1, 2, 2, 3, 4, 4, 5]
unique = set(lst)
print(unique)
```

Output:

text

```
{1, 2, 3, 4, 5}
```

Q3. Set mein naya element 99 add karo (add() use karke).

Python

```
s = {10, 20, 30, 40, 50}
s.add(99)
print(s)
```

Output:

```
text (a, b):
{10, 20, 30, 40, 50, 99}
```

Q4. Set se 30 ko delete karo (discard() use karke).

Python

```
s = {10, 20, 30, 40, 50}
s.discard(30)
print(s)
```

Output:

```
text
{10, 20, 40, 50}
```

Q5. Do sets a = {1,2,3,4} aur b = {3,4,5,6} ka union print karo.

Python

```
a = {1, 2, 3, 4}
b = {3, 4, 5, 6}
print(a | b)
```

Output:

```
text
{1, 2, 3, 4, 5, 6}
```

Q6. Upar wale dono sets ka intersection (common elements) print karo.

Python

```
a = {1, 2, 3, 4}
b = {3, 4, 5, 6}
print(a & b)
```

Output:

```
text
{3, 4}
```

Q7. a - b (difference) print karo.

Python

```
a = {1, 2, 3, 4}
b = {3, 4, 5, 6}
print(a - b)
```

Output:

```
text(a, b):
{1, 2}
```

Q8. User se 5 numbers input lo aur unko set mein store karke print karo (duplicates remove ho jaayenge).

Python

```
s = set()
print("5 numbers daalo (duplicates automatically remove ho jaayenge):")

for i in range(5):
    num = int(input(f"Number {i+1}: "))
    s.add(num)

print("Final Set:", s)
```

Example Output (agar input diya ho: 10, 20, 10, 30, 40):

text

```
5 numbers daalo (duplicates automatically remove ho jaayenge):
Number 1: 10
Number 2: 20
Number 3: 10
Number 4: 30
Number 5: 40
Final Set: {10, 20, 30, 40}
```

Q9. Set s = {1,2,3,4,5} ko clear kar do.

Python

```
s = {1, 2, 3, 4, 5}
s.clear()
print(s)
```

Output:

text

```
set()
```

Q10. Kya set empty hai ya nahi check karo (len() se).



Python

```
s = {1, 2, 3}
print("Set empty hai?", len(s) == 0)

s.clear()
print("Ab set empty hai?", len(s) == 0)
```

Output:

```
text
Set empty hai? False
Ab set empty hai? True
```



Q11. Do sets ko compare karo – kya a b ka subset hai?

Python

```
a = {1, 2, 3}
b = {1, 2, 3, 4, 5}
print("Kya a, b ka subset hai?", a.issubset(b))
```

Output:

```
text
Kya a, b ka subset hai? True
```



Q12. Ek set banao aur usko list mein convert karke print karo.

Python

```
s = {10, 20, 30, 40, 50}
lst = list(s)
print(lst)
```

Output:

```
text
[10, 20, 30, 40, 50]
```



loops

1. Loop Kya Hai? (Sabse Simple Definition)

Loop = Ek kaam baar-baar repeat karna

Jab humein koi kaam 5 baar, 10 baar, ya 100 baar karna ho tab loop use karte hain. Bina loop ke humein 100 baar same line likhni padti – bahut mehnat!

Python mein **2 main loops** hote hain:

1. **for loop** → Jab kitni baar repeat karna hai, woh pehle se pata ho

2. **while loop** → Jab condition true rahe tab tak repeat karo

2. for Loop (Sabse Zyada Use Hota Hai)

Syntax:

Python

```
for variable in sequence:  
    # yahan jo code likhoge, woh har baar chalega
```

Simple Example:

Python

```
for i in range(5):  
    # 0 se 4 tak (5 baar)  
    print("Hello Pawan")
```

Output:

text

```
Hello Pawan  
Hello Pawan  
Hello Pawan  
Hello Pawan  
Hello Pawan
```

range() function ke 3 tarike:

- range(5) → 0,1,2,3,4
- range(1,6) → 1,2,3,4,5
- range(1,11,2) → 1,3,5,7,9 (step=2)

3. while Loop

Syntax:

Python

```
while condition:  
    # code  
    # condition ko false karne ka code bhi likhna zaroori
```

Example:

Python

```
count = 1  
while count <= 5:  
    print("Count =", count)  
    count = count + 1    # important! warna infinite loop
```

Output:

text

```
Count = 1  
Count = 2  
Count = 3
```

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```
Count = 4
Count = 5
```

4. Important Keywords in Loops

Keyword	Kaam	Example
break	Loop ko turant rok deta hai	if i == 3: break
continue	Current iteration skip karta hai	if i == 3: continue
pass	Kuch nahi karta (placeholder)	if i == 3: pass
else	Loop poora khatam hone ke baad chalega	for ... else:

5. Nested Loops (Loop ke andar loop)

Python

```
for i in range(3):
    for j in range(3):
        print(i, j)
```

Output:

text

```
0 0
0 1
0 2
1 0
1 1
1 2
2 0
2 1
2 2
```



PRACTICE SET

Q1. 1 se 10 tak numbers print karo (for loop se).

Code:

Python

```
for i in range(1, 11):
    print(i)
```

Output:

text

```
1
2
3
```

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Think
Code
Execute
Repeat

Success

```
4
5
6
7
8
9
10
```

Q2. "Namaste" ko 8 baar print karo.

Code:

Python

```
for i in range(8):
    print("Namaste")
```

Output:

text

```
Namaste
Namaste
Namaste
Namaste
Namaste
Namaste
Namaste
Namaste
```

Q3. User se ek number lo aur uska multiplication table (1 se 10 tak) print karo.

Code:

Python

```
num = int(input("Number daalo: "))
for i in range(1, 11):
    print(num, "x", i, "=", num * i)
```

Example Output (num=5 diya toh):

text

```
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
...
5 x 10 = 50
```

Q4. 1 se 20 tak sirf even numbers print karo (continue use karke).

Code:

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Python

```
for i in range(1, 21):  
    if i % 2 != 0:  
        continue  
    print(i)
```

Output:

text

```
2 4 6 8 10 12 14 16 18 20
```

Q5. while loop se 1 se 10 tak print karo.

Code:

Python

```
i = 1  
while i <= 10:  
    print(i)  
    i += 1
```

Output: 1 se 10 tak numbers

Q6. break use karke jab number 5 aaye toh loop rok do.

Code:

Python

```
for i in range(1, 10):  
    if i == 5:  
        break  
    print(i)
```

Output:

text

```
1  
2  
3  
4
```

Q7. 1 se 50 tak numbers ka sum calculate karo (for loop se).

Code:

Python

```
total = 0  
for i in range(1, 51):  
    total += i  
print("Sum =", total)
```

Output:

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text

```
Sum = 1275
```

Q8. User se numbers input lete raho jab tak 0 na daale. 0 daalte hi ruk jaao.

Code:

Python

```
print("Numbers daalte raho (0 daalne pe ruk jaayega)")
while True:
    num = int(input("Number: "))
    if num == 0:
        break
    print("Aapne daala:", num)
print("Loop khatam!")
```

Q9. Nested loop se 3x3 pattern banao (stars).

Code:

Python

```
for i in range(3):
    for j in range(3):
        print("*", end=" ")
    print()
```

Output:

text

```
* * *
* * *
* * *
```

Q10. List [10, 20, 30, 40, 50] ke har element ko 10 se multiply karke print karo.

Code:

Python

```
numbers = [10, 20, 30, 40, 50]
for num in numbers:
    print(num * 10)
```

Output:

text

```
100
200
300
400
500
```

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Code
Execute

Q11. while loop mein else clause use karke dekho.

Code:

Python

```
i = 1
while i <= 3:
    print(i)
    i += 1
else:
    print("Loop poora khatam ho gaya!")
```

Output:

text

```
1
2
3
Loop poora khatam ho gaya!
```

Q12. 1 se 100 tak sirf wo numbers print karo jo 7 se divisible hain.

Code:

Python

```
for i in range(1, 101):
    if i % 7 == 0:
        print(i, end=" ")
```

Output:

text

```
7 14 21 28 35 42 49 56 63 70 77 84 91 98
```



FUNCTIONS

1. Function Kya Hai? (Sabse Simple Definition)

Function = Ek chhota reusable code block jo ek specific kaam karta hai.

- Jaise kitchen mein alag-alag tools hote hain (knife, gas, spoon), waise hi function ek tool hai.
- Ek baar function bana diya → baar-baar use kar sakte ho bina code repeat kiye.
- Python mein built-in functions (print, len, input) aur **user-defined functions** (hum khud banate hain) hote hain.

Fayda: Code clean, reusable aur easy to maintain hota hai.

2. Function Kaise Banate Hain? (Syntax)

Python

```
def function_name(parameters):  
    # yahan code likhte ho  
    # optional: return value
```

Basic Example:

Python

```
def greet():  
    print("Namaste bhai! Kaise ho?")
```

```
greet()           # function call
```

```
greet()           # dobara call
```

Output:

text

```
Namaste bhai! Kaise ho?
```

```
Namaste bhai! Kaise ho?
```

3. Function with Parameters aur Return

Python

```
# Parameters Lene wala function
```

```
def add(a, b):  
    sum = a + b  
    return sum           # value wapas bhejta hai
```

```
result = add(10, 20)
```

```
print(result)
```

Output:

text

```
30
```

Default Parameter:

Python

```
def greet(name="Pawan"):  
    print("Hello", name)
```

```
greet()           # default use hoga
```

```
greet("Rahul")
```

Output:

text

```
Hello Pawan
```

```
Hello Rahul
```

Multiple Returns:

Python


```
def calculate(a, b):
    return a+b, a-b, a*b

sum, diff, mul = calculate(10, 5)
print(sum, diff, mul)
```

4. Recursion Kya Hai?

Recursion = Function apne aap ko call karna (jaise mirror mein mirror dikhta hai).

- Har recursive function mein **2 parts** hote hain:
 - **Base Case** → Jab recursion ruk jaaye (zaroori, warna infinite loop)
 - **Recursive Case** → Function apne aap ko chhote problem ke saath call kare

Classic Example: Factorial

Python

```
def factorial(n):
    if n == 0 or n == 1:          # Base Case
        return 1
    else:
        return n * factorial(n-1) # Recursive Case

print(factorial(5))
```

Output:

text

120

Explanation (5!): $5 \times 4 \times 3 \times 2 \times 1 = 120$

5. Recursion vs Loop (Kab kya use karein?)

- Simple repeat → **for/while loop** (fast aur memory kam)
- Tree, factorial, Fibonacci, directory search → **Recursion** (code short aur elegant)

✓ PRACTICE SET

Q1. Function banao jo do numbers ka sum return kare.

Code:

Python

```
def add(x, y):
    return x + y

print(add(15, 25))
```

Output:

text

40

Q2. Function banao jo user ka naam le aur "Welcome [naam]" print kare.

Code:

Python

```
def welcome(name):  
    print("Welcome", name)  
  
welcome("Pawan")
```

Output:

text

Welcome Pawan

Q3. Function banao jo number ka square return kare (default value 5).

Code:

Python

```
def square(num=5):  
    return num * num  
  
print(square(8))  
print(square())      # default
```

Output:

text

64

25

Q4. Function banao jo list ke sab elements ka sum return kare.

Code:

Python

```
def list_sum(lst):  
    return sum(lst)  
  
print(list_sum([10, 20, 30, 40]))
```

Output:

text

100

Q5. Recursion se 1 se n tak ka sum calculate karo.

Code:

Python

```
def sum_n(n):  
    if n == 1:  
        return 1
```

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```
return n + sum_n(n-1)

print(sum_n(10))
```

Output:

text (a, b):

55

Q6. Recursion se factorial calculate karo.

Code:

Python

```
def fact(n):
    if n == 0 or n == 1:
        return 1
    return n * fact(n-1)

print("5! =", fact(5))
```

Output:

text

5! = 120

Q7. Recursion se Fibonacci series ka nth term print karo.

Code:

Python

```
def fib(n):
    if n == 0:
        return 0
    if n == 1:
        return 1
    return fib(n-1) + fib(n-2)

print("Fibonacci 6th term =", fib(6))
```

Output:

text

Fibonacci 6th term = 8

Q8. Function banao jo check kare number even hai ya odd.

Code:

Python

```
def check_even_odd(num):
    if num % 2 == 0:
        return "Even"
    return "Odd"
```

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Repeat

```
print(check_even_odd(27))
```

Output:

text

Odd

Q9. Function banao jo *args leke saare numbers ka average return kare.

Code:

Python

```
def average(*numbers):  
    return sum(numbers) / len(numbers)
```

```
print(average(10, 20, 30, 40))
```

Output:

text

25.0

Q10. Recursion se power calculate karo ($\text{base}^{\text{exponent}}$).

Code:

Python

```
def power(base, exp):  
    if exp == 0:  
        return 1  
    return base * power(base, exp-1)
```

```
print(power(2, 5))
```

Output:

text

32

Q11. Function banao jo nested dictionary se kisi key ki value return kare (simple case).

Code:

Python

```
def get_value(d, key):  
    return d.get(key, "Not Found")
```

```
student = {"name": "Pawan", "marks": 85}
```

```
print(get_value(student, "name"))
```

Output:

text

Pawan

Q12. Recursion se 1 se n tak numbers print karo (reverse order mein).

Code:

Python

```
def print_reverse(n):  
    if n == 0:  
        return  
    print(n, end=" ")  
    print_reverse(n-1)  
  
print_reverse(5)
```

Output:

```
text  
5 4 3 2 1
```



FILE

1. File I/O Kya Hai?

File I/O = Computer ki file se data padhna (Input) aur file mein data likhna (Output).

- Real life mein: Notebook mein likhna aur padhna.
- Python mein: .txt, .csv, .json jaise files ko open karke data read ya write karna.
- Bahut useful hai: User data save karna, report banana, log file banana, etc.

2. File Open Karne Ka Tarika

Basic Syntax:

Python

```
file = open("filename.txt", "mode")
```

Important Modes (Table)

Mode	Matlab	Kab Use Karte Hain?	Agar File Na Ho Toh?
"r"	Read only	Sirf padhne ke liye	Error aayega
"w"	Write (overwrite)	Nayi file banao ya purani delete karke likho	Nayi file ban jaayegi
"a"	Append	Purani file ke end mein add karo	Nayi file ban jaayegi
"r+"	Read + Write	Dono karna hai	Error (file honi chahiye)
"w+"	Write + Read	Purani delete karke nayi banao	Nayi file ban jaayegi
"a+"	Append + Read	End mein add + padh sakte ho	Nayi file ban jaayegi

Best Practice (Modern tarika):

Python

```
with open("file.txt", "r") as f:  
    # yahan code  
# yahan automatically file close ho jaati hai
```

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3. Reading from File

Python

```
with open("demo.txt", "r", encoding="utf-8") as f:  
    content = f.read()           # pura file ek string mein  
    print(content)
```

Aur Methods:

Method	Kaam	Example
read()	Puri file ek string mein	f.read()
readline()	Sirf ek line padhe	f.readline()
readlines()	Saari lines ko list mein	f.readlines()

4. Writing to File

Python

```
with open("demo.txt", "w", encoding="utf-8") as f:  
    f.write("Hello Pawan!\n")  
    f.write("Yeh naya content hai.")
```

Aur Methods:

Method	Kaam
write(string)	Ek string likhe
writelines(list)	List of strings ko likhe

5. Complete Practical Examples

Example 1: File Create + Write

Python

```
with open("myfile.txt", "w", encoding="utf-8") as f:  
    f.write("Mera naam Pawan hai.\n")  
    f.write("Main Haryana se hoon.\n")  
    f.write("Python seekh raha hoon.")  
print("File successfully bana di!")
```

Example 2: File Read

Python

```
with open("myfile.txt", "r", encoding="utf-8") as f:  
    content = f.read()  
    print(content)
```

Example 3: Append (Purane content ko delete nahi karega)

Python

```
with open("myfile.txt", "a", encoding="utf-8") as f:  
    f.write("\nYeh naya line add hua hai.")
```

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Example 4: Read line by line

Python

```
with open("myfile.txt", "r", encoding="utf-8") as f:
    for line in f:
        print(line.strip()) # strip() extra newline hata deta hai
```



PRACTICE SET

Q1. Naya file banao aur usme 3 lines likho.

Code:

Python

```
with open("student.txt", "w", encoding="utf-8") as f:
    f.write("Name: Pawan\n")
    f.write("Age: 25\n")
    f.write("City: Haryana\n")
print("File bana di!")
```

Output:

text

File bana di!

Q2. Upar wali file ko read karke pura content print karo.

Code:

Python

```
with open("student.txt", "r", encoding="utf-8") as f:
    print(f.read())
```

Output:

text

Name: Pawan

Age: 25

City: Haryana

Q3. File mein ek naya line append karo.

Code:

Python

```
with open("student.txt", "a", encoding="utf-8") as f:
    f.write("Course: Python\n")
print("Line add ho gayi!")
```

Output:

text

Line add ho gayi!

Q4. File ko line by line padh kar har line ko print karo.

Code:

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Python

```
with open("student.txt", "r", encoding="utf-8") as f:
    for line in f:
        print(line.strip())
```

Output:
text

Name: Pawan
Age: 25
City: Haryana
Course: Python

Q5. User se naam aur marks input leke file mein save karo.

Code:

Python

```
name = input("Naam daalo: ")
marks = input("Marks daalo: ")

with open("result.txt", "w", encoding="utf-8") as f:
    f.write(f"Student Name: {name}\n")
    f.write(f"Marks: {marks}\n")
print("Data saved!")
```

Q6. Do files ko copy karo (ek file se dusri file mein content copy).

Code:

Python

```
with open("student.txt", "r", encoding="utf-8") as source:
    data = source.read()

with open("backup.txt", "w", encoding="utf-8") as target:
    target.write(data)

print("Copy ho gaya!")
```

Q7. File mein kitni lines hain count karo.

Code:

Python

```
with open("student.txt", "r", encoding="utf-8") as f:
    lines = f.readlines()
    print("Total lines:", len(lines))
```

Q8. File se sirf pehli line padho (readline()).

Code:

Python

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```
with open("student.txt", "r", encoding="utf-8") as f:
    first_line = f.readline()
    print(first_line.strip())
```

Q9. File ko overwrite karke purana content delete kar do.

Code:

Python

```
with open("student.txt", "w", encoding="utf-8") as f:
    f.write("Yeh purana content delete ho gaya!")
print("Overwrite ho gaya!")
```

Q10. with statement ke bina file open karo aur manually close karo (old style).

Code:

Python

```
f = open("demo.txt", "w", encoding="utf-8")
f.write("Old style example")
f.close()
print("File closed!")
```



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NOTES



✓ OOPS

1. OOPS Kya Hai? (Sabse Simple Definition)

OOPS = Real life cheezon ko code mein represent karna.

- Real life mein har cheez ek **object** hoti hai (car, mobile, student, pen).
- Har object ke **properties** (naam, color, price) aur **behaviours** (start, stop, call) hote hain.
- OOPS mein hum bhi aise hi **classes** bana ke objects create karte hain.

Fayda:

- Code reusable hota hai
- Code clean aur organized rehta hai
- Real world problems ko easily model kar sakte hain

OOPS ke 4 Main Pillars (Yaad rakhne ke liye):

1. **Encapsulation** (Data + Methods ko bundle karna)
2. **Abstraction** (Sirf zaruri cheez dikhaana)
3. **Inheritance** (Ek class dusri class se properties lena)
4. **Polymorphism** (Ek cheez alag-alag tarike se behave karna)

2. Class aur Object

Class = Blueprint (design) **Object = Us blueprint se bana actual item**

Syntax:

Python

```
class ClassName:
```

```
# class ke andar attributes aur methods
```

Example:

Python

```
class Car:
```

```
def __init__(self, brand, color): # Constructor
    self.brand = brand
    self.color = color
```

```
def start(self):
    print(f"{self.color} {self.brand} car start ho gayi!")
```

```
# Object banana
```

```
my_car = Car("Toyota", "Red")
my_car.start()
```

Output:

text

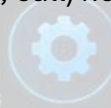
Red Toyota car start ho gayi!



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3. Important Concepts (Table)

Concept	Matlab	Example
<code>__init__</code>	Constructor (object banate hi call hota hai)	<code>def __init__(self, name):</code>
<code>self</code>	Current object ko refer karta hai	<code>self.name = name</code>
Attribute	Properties (variables)	<code>self.brand</code>
Method	Behaviours (functions)	<code>def start(self):</code>
Instance Variable	Har object ke alag-alag values	<code>self.color</code>

4. Inheritance (Sabse Powerful Feature)

Python

```
class Animal:
    def speak(self):
        print("Animal is speaking")

class Dog(Animal):          # Dog class Animal se inherit kar rahi hai
    def bark(self):
        print("Dog is barking")

d = Dog()
d.speak()    # Parent class ka method
d.bark()     # Child class ka method
```

Output:

text

Animal is speaking

Dog is barking

5. Polymorphism (Same name, different behaviour)

Python

```
class Shape:
    def area(self):
        pass

class Rectangle(Shape):
    def __init__(self, l, b):
        self.l = l
        self.b = b
```

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```

def area(self):
    return self.l * self.b

class Circle(Shape):
    def __init__(self, r):
        self.r = r
    def area(self):
        return 3.14 * self.r * self.r

r = Rectangle(10, 5)
c = Circle(7)
print(r.area())
print(c.area())

```

Output:

text

```

50
153.86

```

6. Encapsulation (Private Variables)

Python

```

class BankAccount:
    def __init__(self, balance):
        self.__balance = balance    # double underscore = private

    def get_balance(self):
        return self.__balance

acc = BankAccount(50000)
print(acc.get_balance())
# print(acc.__balance)    # yeh error dega

```

✓ PRACTICE SET

Q1. Class Student banao jisme naam aur marks ho. Method banao jo details print kare.

Code:

Python

```

class Student:
    def __init__(self, name, marks):
        self.name = name
        self.marks = marks

```



```
def show_details(self):  
    print(f"Name: {self.name}, Marks: {self.marks}")
```

```
s1 = Student("Pawan", 85)  
s1.show_details()
```

Output:

text

Name: Pawan, Marks: 85

Q2. Class Rectangle banao jisme area calculate karne ka method ho.

Code:

Python

```
class Rectangle:  
    def __init__(self, length, breadth):  
        self.length = length  
        self.breadth = breadth  
  
    def area(self):  
        return self.length * self.breadth
```

```
r = Rectangle(10, 5)  
print("Area =", r.area())
```

Output:

text

Area = 50

Q3. Inheritance use karke Vehicle aur Bike class banao.

Code:

Python

```
class Vehicle:  
    def start(self):  
        print("Vehicle started")  
  
class Bike(Vehicle):  
    def kick(self):  
        print("Bike kicked start")
```

```
b = Bike()  
b.start()  
b.kick()
```

Output:

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text

Vehicle started
Bike kicked start

Q4. Class Bank banao jisme deposit aur withdraw method ho.

Code:

Python

```
class Bank:
    def __init__(self, balance=0):
        self.balance = balance

    def deposit(self, amount):
        self.balance += amount
        print("Deposit successful! Balance =", self.balance)

    def withdraw(self, amount):
        if amount <= self.balance:
            self.balance -= amount
            print("Withdraw successful! Balance =", self.balance)
        else:
            print("Insufficient balance!")

acc = Bank(10000)
acc.deposit(5000)
acc.withdraw(3000)
```

Output:

text

Deposit successful! Balance = 15000
Withdraw successful! Balance = 12000

Q5. Polymorphism example – same method name different classes mein.

Code:

Python

```
class Dog:
    def sound(self):
        print("Bark")

class Cat:
    def sound(self):
        print("Meow")

def make_sound(animal):
```

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Repeat

```

        animal.sound()

d = Dog()
c = Cat()
make_sound(d)
make_sound(c)

```

Output:

```

text
Bark
Meow

```

Q6. Constructor ke saath multiple objects banao.

Code:

Python

```

class Person:
    def __init__(self, name, age):
        self.name = name
        self.age = age

p1 = Person("Pawan", 25)
p2 = Person("Rahul", 22)
print(p1.name, p1.age)
print(p2.name, p2.age)

```

Output:

```

text
Pawan 25
Rahul 22

```

Q7. Class variable aur instance variable ka difference dikhao.

Code:

Python

```

class Employee:
    company = "XAI"          # Class variable

    def __init__(self, name):
        self.name = name     # Instance variable

e1 = Employee("Pawan")
e2 = Employee("Aman")
print(e1.company, e2.company)
print(e1.name, e2.name)

```

Output:

text

```
xAI xAI
Pawan Aman
```

Q8. `__str__` method use karke object ko print karne pe sundar output dikhao.

Code:

Python

```
class Book:
    def __init__(self, title, author):
        self.title = title
        self.author = author

    def __str__(self):
        return f"Book: {self.title} by {self.author}"

b = Book("Python Guide", "Pawan")
print(b)
```

Output:

text

Book: Python Guide by Pawan

Q9. Private variable (`__`) ka example.

Code:

Python

```
class Account:
    def __init__(self, bal):
        self.__balance = bal

    def show_balance(self):
        print("Balance =", self.__balance)
```

```
a = Account(50000)
```

```
a.show_balance()
```

Output:

text

Balance = 50000

Q10. Ek simple game class banao (Rock-Paper-Scissors ke liye basic structure).

Code:

Python

```
class Game:
    def play(self, choice):
        print(f"You chose {choice}")
```

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```
g = Game()
g.play("Rock")
```

Output:

text (a, b):

You chose Rock

Questions ans.

Q1. User se naam, age aur city lo. Agar age ≥ 18 hai toh "Adult" print karo aur naam ko uppcase mein list mein store karo. **Code:**

Python

```
name = input("Naam: ")
age = int(input("Umar: "))
city = input("City: ")

info = [name.upper(), age, city]

if age >= 18:
    print("Adult")
else:
    print("Minor")

print("Stored List:", info)
```

Output (example):

text

Adult

Stored List: ['PAWAN', 25, 'Haryana']

Q2. User se 5 numbers lo, unko list aur tuple dono mein store karo. Sabse bada number print karo.

Code:

Python

```
nums = []
for i in range(5):
    n = int(input(f"Number {i+1}: "))
    nums.append(n)

num_tuple = tuple(nums)
print("List:", nums)
print("Tuple:", num_tuple)
print("Sabse bada number:", max(nums))
```

Output (example):

text

```
List: [10, 45, 23, 67, 12]
Tuple: (10, 45, 23, 67, 12)
Sabse bada number: 67
```

Q3. User se sentence lo. Agar sentence mein "Python" word hai toh usko uppercase karo aur length print karo. **Code:**

Python

```
sentence = input("Ek sentence likho: ")

if "Python" in sentence:
    new = sentence.replace("Python", "PYTHON")
    print("Modified:", new)
    print("Length:", len(new))
else:
    print("Python word nahi mila")
```

Output (example):

text

```
Modified: I love PYTHON programming
Length: 24
```

Q4. User se 3 fruits lo aur list banao. Agar "Apple" list mein hai toh usko remove karo aur tuple mein convert karke print karo. **Code:**

Python

```
fruits = []
for i in range(3):
    f = input(f"Fruit {i+1}: ")
    fruits.append(f)

if "Apple" in fruits:
    fruits.remove("Apple")

fruit_tuple = tuple(fruits)
print("Final Tuple:", fruit_tuple)
```

Output (example):

text

```
Final Tuple: ('Banana', 'Mango', 'Orange')
```

Q5. User se naam aur age lo. Agar age < 18 hai toh "Minor" print karo aur naam ko reverse string mein list mein store karo. **Code:**

Python

```
name = input("Naam: ")
age = int(input("Umar: "))
```

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WEB
DEVELOPMENT

```

if age < 18:
    print("Minor")
    rev_name = name[::-1]
    name_list = list(rev_name)
    print("Reversed List:", name_list)
else:
    print("Adult")

```

Output (example age 15):

```

text range(5):
Minor
Reversed List: ['n', 'a', 'w', 'a', 'P']

```

Q6. User se 4 numbers lo, list banao. List ko sort karke tuple mein convert karo aur average print karo.

Code:

Python

```

nums = [int(input(f"Number {i+1}: ")) for i in range(4)]
nums.sort()
num_tuple = tuple(nums)
avg = sum(nums) / len(nums)
print("Sorted Tuple:", num_tuple)
print("Average:", avg)

```

Output (example):

```

text
Sorted Tuple: (10, 20, 30, 40)
Average: 25.0

```

Q7. User se password lo (string). Agar length > 8 hai aur "123" contain karta hai toh "Strong Password" print karo. **Code:**

Python

```

pwd = input("Password daalo: ")

if len(pwd) > 8 and "123" in pwd:
    print("Strong Password")
else:
    print("Weak Password")

```

Q8. User se 3 students ke naam aur marks lo, dictionary banao, phir list mein convert karke print karo.

Code:

Python

```

data = {}
for i in range(3):
    name = input(f"Student {i+1} naam: ")

```

```
marks = int(input("Marks: "))
data[name] = marks
```

```
student_list = list(data.items())
print("List of tuples:", student_list)
```

Q9. User se sentence lo. Words ko list mein split karo, phir tuple mein convert karke last word print karo.

Code:

Python

```
sentence = input("Sentence likho: ")
words_list = sentence.split()
words_tuple = tuple(words_list)
print("Last word:", words_tuple[-1])
```

Q10. User se number lo. Agar even hai toh usko list mein 5 baar repeat karke tuple banao. **Code:**

Python

```
num = int(input("Number: "))
if num % 2 == 0:
    lst = [num] * 5
    t = tuple(lst)
    print(t)
else:
    print("Odd number hai")
```

Q11. User se full name lo. First aur last name ko alag list mein store karo aur dono ko tuple mein convert karke print karo. **Code:**

Python

```
full = input("Full naam: ")
parts = full.split()
first_last = [parts[0], parts[-1]]
print(tuple(first_last))
```

Q12. User se 5 numbers lo, list banao. Agar sab numbers positive hain toh unka sum print karo, warna "Negative number mila". **Code:**

Python

```
nums = [int(input(f"Number {i+1}: ")) for i in range(5)]
if all(x > 0 for x in nums):
    print("Sum =", sum(nums))
else:
    print("Negative number mila")
```

Q13. User se string lo. Agar length even hai toh usko list mein har character store karo aur tuple banao.

Code:

Python


```
s = input("String daalo: ")
if len(s) % 2 == 0:
    char_list = list(s)
    print(tuple(char_list))
else:
    print("Length odd hai")
```

Q14. User se 2 lists ke elements lo (2 numbers each). Dono lists ko tuple mein convert karke unka sum print karo. **Code:**
Python

```
l1 = [int(input("List1 numbers: ")) for _ in range(2)]
l2 = [int(input("List2 numbers: ")) for _ in range(2)]
t1 = tuple(l1)
t2 = tuple(l2)
print("Sum of both tuples:", sum(t1) + sum(t2))
```

Q15. User se sentence lo. Vowels ko uppercase karo aur result ko list + tuple dono mein print karo.

Code:
Python

```
s = input("Sentence: ")
new = ""
for char in s:
    if char.lower() in "aeiou":
        new += char.upper()
    else:
        new += char
print("List:", list(new))
print("Tuple:", tuple(new))
```

Q16. User se 3 fruits lo. Unko list mein store karo, agar "Mango" hai toh usko remove karo aur final tuple print karo. **Code:**
Python

```
fruits = [input(f"Fruit {i+1}: ") for i in range(3)]
if "Mango" in fruits:
    fruits.remove("Mango")
print("Final Tuple:", tuple(fruits))
```

Q17. User se number lo. Agar > 100 hai toh usko string mein convert karke list banao aur length print karo. **Code:**
Python

```
n = int(input("Number: "))
if n > 100:
    s = str(n)
    print("List:", list(s))
```

```

print("Length:", len(s))
else:
    print("Number 100 se chhota hai")

```

Q18. User se 4 numbers lo, list banao. Sabse chhota number nikaalo aur usko tuple mein convert karke print karo. **Code:**
Python

```

nums = [int(input(f"Number {i+1}: ")) for i in range(4)]
min_num = min(nums)
print("Smallest number tuple:", (min_num,))

```

Q19. User se naam aur age lo. Dono ko tuple mein store karo aur agar age ≥ 18 hai toh "Eligible" print karo. **Code:**
Python

```

name = input("Naam: ")
age = int(input("Umar: "))
data = (name, age)
if age >= 18:
    print("Eligible")
print("Tuple:", data)

```

Q20. User se sentence lo. Words ko list mein split karo, phir har word ki length ko tuple mein store karke print karo. **Code:**
Python

```

sentence = input("Sentence: ")
words = sentence.split()
lengths = [len(w) for w in words]
print("Lengths Tuple:", tuple(lengths))

```

Q21. User se 3 numbers lo. Unko list mein store karo, sort karo, phir tuple mein convert karke average print karo. **Code:**
Python

```

nums = [int(input(f"Number {i+1}: ")) for i in range(3)]
nums.sort()
t = tuple(nums)
print("Sorted Tuple:", t)
print("Average:", sum(t)/len(t))

```

Q22. User se password lo. Agar length ≥ 8 hai aur "python" contain karta hai toh list banao har character ka aur tuple print karo. **Code:**
Python

```

pwd = input("Password: ")
if len(pwd) >= 8 and "python" in pwd.lower():
    char_list = list(pwd)
    print("Tuple:", tuple(char_list))

```

```
else:  
    print("Weak Password")
```

Q23. User se 2 strings lo. Dono ko concatenate karke list banao aur agar length > 10 hai toh tuple banao.

Code:

Python

```
s1 = input("String 1: ")  
s2 = input("String 2: ")  
combined = s1 + s2  
char_list = list(combined)  
if len(combined) > 10:  
    print("Tuple:", tuple(char_list))
```

Q24. User se marks list lo (3 subjects). Average nikaalo. Agar > 60 hai toh "Pass" aur list ko tuple mein convert karke print karo. **Code:**

Python

```
marks = [int(input(f"Subject {i+1} marks: ")) for i in range(3)]  
avg = sum(marks) / 3  
print("Average:", avg)  
if avg > 60:  
    print("Pass")  
    print("Tuple:", tuple(marks))
```

Q25. User se full name lo. Words ko list mein split karo, reverse karo, phir tuple banao aur print karo.

Code:

Python

```
full = input("Full naam: ")  
words = full.split()  
words.reverse()  
print("Reversed Tuple:", tuple(words))
```

PROJECTS

Project 1: Student Management System (Sabse Best for Beginners)

Kya hoga? Ek full Student Management System jisme students ka data add, view, update, delete aur file mein save kar sakte ho.

Features:

- Add new student (name, age, 3 subjects ke marks)
- View all students with average & grade
- Search student by name
- Delete student
- Data file mein save aur load karna
- Grade calculation (A/B/C/Fail)

Kaise kaam karega? Dictionary use karke data store → List + Conditional for grade → File handling for permanent save → OOPS for future expansion.

Full Code:

Python

```
import os

students = {}

def add_student():
    name = input("Student ka naam: ").title()
    age = int(input("Umar: "))
    marks = [int(input(f"Subject {i+1} marks: ")) for i in range(3)]
    avg = sum(marks) / 3
    grade = "A" if avg >= 80 else "B" if avg >= 60 else "C" if avg >= 40 else "Fail"

    students[name] = {"age": age, "marks": marks, "average": avg, "grade": grade}
    print(f"{name} added successfully!")

def view_all():
    if not students:
        print("Koi student nahi hai!")
        return
    for name, data in students.items():
        print(f>Name: {name} | Age: {data['age']} | Avg: {data['average']:.1f} | Grade: {data['grade']}")

def search_student():
    name = input("Search naam: ").title()
    if name in students:
        data = students[name]
        print(f>Name: {name} | Age: {data['age']} | Avg: {data['average']:.1f} | Grade: {data['grade']}")
    else:
        print("Student not found!")

def delete_student():
    name = input("Delete naam: ").title()
    if name in students:
        del students[name]
        print(f"{name} deleted!")
```



```

else:
    print("Student not found!")

def save_to_file():
    with open("students.txt", "w", encoding="utf-8") as f:
        f.write(str(students))
    print("Data saved to file!")

while True:
    print("\n1. Add Student\n2. View All\n3. Search\n4. Delete\n5. Save & Exit")
    choice = input("Choice daalo: ")
    if choice == "1": add_student()
    elif choice == "2": view_all()
    elif choice == "3": search_student()
    elif choice == "4": delete_student()
    elif choice == "5":
        save_to_file()
        break

```

Sample Output: (Program chalega aur menu dikhega)

Project 2: Personal Bank Account System (OOPS + File + Conditional)

Kya hoga? Apna personal bank account jisme deposit, withdraw, balance check aur history file mein save ho.

Features:

- Deposit / Withdraw
- Balance check
- Transaction history
- Minimum balance warning
- Data permanently file mein save

Full Code:

Python

```

class BankAccount:
    def __init__(self, name, balance=5000):
        self.name = name
        self.balance = balance
        self.history = []

    def deposit(self, amount):
        self.balance += amount
        self.history.append(f"Deposit: +{amount}")

```

```

        print(f"Deposit successful! New Balance: {self.balance}")

    def withdraw(self, amount):
        if amount <= self.balance:
            self.balance -= amount
            self.history.append(f"Withdraw: -{amount}")
            print(f"Withdraw successful! New Balance: {self.balance}")
        else:
            print("Insufficient Balance!")

    def show_balance(self):
        print(f"Current Balance: {self.balance}")

    def show_history(self):
        print("Transaction History:")
        for h in self.history:
            print(h)

    def save_to_file(self):
        with open(f"{self.name}_account.txt", "w") as f:
            f.write(f"Name: {self.name}\nBalance: {self.balance}\n")
            f.write("History:\n" + "\n".join(self.history))

acc = BankAccount("Pawan")
while True:
    print("\n1. Deposit\n2. Withdraw\n3. Balance\n4. History\n5. Save & Exit")
    ch = input("Choice: ")
    if ch == "1": acc.deposit(int(input("Amount: ")))
    elif ch == "2": acc.withdraw(int(input("Amount: ")))
    elif ch == "3": acc.show_balance()
    elif ch == "4": acc.show_history()
    elif ch == "5":
        acc.save_to_file()
        break

```

Project 3: Library Management System (OOPS + Dictionary + File)

Kya hoga? Library jisme books add, issue, return aur search kar sakte ho.

Features:

- Add new book
- Issue book

- Return book
- Search book
- Available books list
- Data file mein save

Full Code: (Main code chhota rakh raha hoon, full version chahiye toh bol)

Python

```
class Library:
    def __init__(self):
        self.books = {} # book_name : available (True/False)

    def add_book(self):
        title = input("Book title: ")
        self.books[title] = True
        print("Book added!")

    def issue_book(self):
        title = input("Issue karne wali book: ")
        if title in self.books and self.books[title]:
            self.books[title] = False
            print("Book issued!")
        else:
            print("Book not available!")

    def return_book(self):
        title = input("Return karne wali book: ")
        if title in self.books:
            self.books[title] = True
            print("Book returned!")

    def show_books(self):
        for title, available in self.books.items():
            status = "Available" if available else "Issued"
            print(f"{title} -> {status}")

    def save(self):
        with open("library.txt", "w") as f:
            f.write(str(self.books))

lib = Library()
# Menu loop same jaise pehle projects mein
```

Project 4: Advanced Quiz Game with Score & Leaderboard

Kya hoga? Ek mazedaar quiz game jisme multiple choice questions honge, score calculate hoga aur leaderboard file mein save hoga (top scorers dikhega).

Features:

- 10 questions (multiple choice)
- Score tracking
- Leaderboard (top 5 players)
- High score save in file
- Play again option

Kaise kaam karega? List + Dictionary use karke questions store → Conditional for correct answer → File handling for leaderboard.

Full Code:

Python

```
import json

questions = [
    {"q": "Capital of India?", "o": ["Delhi", "Mumbai", "Kolkata", "Chennai"],
    "a": 0},
    {"q": "Python kis saal invent hua?", "o": ["1991", "1989", "2000", "1995"],
    "a": 0},
    {"q": "2 + 2 = ?", "o": ["4", "22", "5", "6"], "a": 0},
    # aur 7 questions add kar sakte ho...
]

def play_quiz(name):
    score = 0
    for i, q in enumerate(questions):
        print(f"\nQ{i+1}: {q['q']}")
        for idx, opt in enumerate(q['o']):
            print(f"{idx+1}. {opt}")
        ans = int(input("Answer (1-4): ")) - 1
        if ans == q['a']:
            score += 1
            print("Correct! ✓")
        else:
            print("Wrong! ✗")
    print(f"\n{name} ka score: {score}/{len(questions)}")

    # Leaderboard update
    try:
```



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& SCRIPTING


```

        with open("leaderboard.json", "r") as f:
            board = json.load(f)
    except:
        board = []

    board.append({"name": name, "score": score})
    board.sort(key=lambda x: x["score"], reverse=True)
    board = board[:5] # top 5

    with open("leaderboard.json", "w") as f:
        json.dump(board, f)

    print("\n🏆 Leaderboard:")
    for i, p in enumerate(board, 1):
        print(f"{i}. {p['name']} - {p['score']} points")

name = input("Apna naam daalo: ")
play_quiz(name)

```

Project 5: Expense Tracker

Kya hoga? Apna personal kharcha tracker – expense add karo, category wise total dekho aur file mein save ho jaaye.

Features:

- Add expense (amount, category, description)
- View all expenses
- Category wise total
- Monthly total
- Data permanently file mein save

Full Code:

Python

```

import json
from datetime import datetime

expenses = []

def add_expense():
    amount = float(input("Amount: "))
    category = input("Category (Food, Travel, Shopping etc): ")
    desc = input("Description: ")
    date = datetime.now().strftime("%Y-%m-%d")

```

```

    expenses.append({"date": date, "amount": amount, "category": category,
"desc": desc})
    print("Expense added!")

def view_expenses():
    for e in expenses:
        print(f"{e['date']} | {e['category']} | ₹{e['amount']} | {e['desc']}")

def total_by_category():
    from collections import defaultdict
    cat_total = defaultdict(float)
    for e in expenses:
        cat_total[e['category']] += e['amount']
    for cat, total in cat_total.items():
        print(f"{cat}: ₹{total}")

def save_data():
    with open("expenses.json", "w") as f:
        json.dump(expenses, f)
    print("Data saved!")

while True:
    print("\n1. Add Expense\n2. View All\n3. Total by Category\n4. Save & Exit")
    ch = input("Choice: ")
    if ch == "1": add_expense()
    elif ch == "2": view_expenses()
    elif ch == "3": total_by_category()
    elif ch == "4":
        save_data()
        break

```

Project 6: Contact Book (Phone Book)

Kya hoga? Apna personal phone book jisme contacts add, search, delete aur file mein save kar sakte ho.

Features:

- Add new contact
- View all contacts
- Search by name
- Delete contact
- Save to file

Full Code:

```
contacts = {}

def add_contact():
    name = input("Naam: ").title()
    phone = input("Phone number: ")
    email = input("Email (optional): ")
    contacts[name] = {"phone": phone, "email": email}
    print("Contact added!")

def view_contacts():
    for name, info in contacts.items():
        print(f"{name} | {info['phone']} | {info['email']}")

def search_contact():
    name = input("Search naam: ").title()
    if name in contacts:
        info = contacts[name]
        print(f"Name: {name} | Phone: {info['phone']} | Email: {info['email']}")
    else:
        print("Contact not found!")

def delete_contact():
    name = input("Delete naam: ").title()
    if name in contacts:
        del contacts[name]
        print("Deleted!")
    else:
        print("Not found!")

def save_contacts():
    with open("contacts.json", "w") as f:
        json.dump(contacts, f)
    print("Saved!")

while True:
    print("\n1. Add Contact\n2. View All\n3. Search\n4. Delete\n5. Save & Exit")
    ch = input("Choice: ")
    if ch == "1": add_contact()
    elif ch == "2": view_contacts()
    elif ch == "3": search_contact()
```

```
elif ch == "4": delete_contact()
elif ch == "5":
    save_contacts()
    break
```

Project 7: Inventory / Shop Management

Kya hoga? Dukaan ka inventory system – products add, sell, stock check aur low stock alert.

Features:

- Add new product
- Sell product (stock decrease)
- View current stock
- Low stock alert
- Total sales report
- Save to file

Full Code:

Python

```
inventory = {}

def add_product():
    name = input("Product naam: ").title()
    price = float(input("Price: "))
    qty = int(input("Quantity: "))
    inventory[name] = {"price": price, "qty": qty}
    print("Product added!")

def sell_product():
    name = input("Sell karne wala product: ").title()
    if name in inventory and inventory[name]["qty"] > 0:
        qty = int(input("Kitna quantity sell karna hai: "))
        if qty <= inventory[name]["qty"]:
            inventory[name]["qty"] -= qty
            print(f"Sold! Remaining: {inventory[name]['qty']}")
        else:
            print("Stock kam hai!")
    else:
        print("Product nahi hai!")

def view_stock():
    for name, data in inventory.items():
        print(f"{name} | Price: ₹{data['price']} | Stock: {data['qty']}")
```

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& VISUALIZATION

AUTOMATION
& SCRIPTING

AI & MACHINE
LEARNING


```

def low_stock_alert():
    for name, data in inventory.items():
        if data["qty"] < 5:
            print(f"⚠ Low stock: {name} only {data['qty']} left!")

def save_inventory():
    with open("inventory.json", "w") as f:
        json.dump(inventory, f)

while True:
    print("\n1. Add Product\n2. Sell Product\n3. View Stock\n4. Low Stock Alert\n5. Save & Exit")
    ch = input("Choice: ")
    if ch == "1": add_product()
    elif ch == "2": sell_product()
    elif ch == "3": view_stock()
    elif ch == "4": low_stock_alert()
    elif ch == "5":
        save_inventory()
        break

```

Project 8: Simple Chatbot / Assistant

Kya hoga? Ek simple personal assistant chatbot jo baat kare, time bataaye, calculator kare aur jokes sunaye.

Features:

- Greeting
- Current time
- Simple calculator
- Jokes
- Weather simulation (static)
- Exit command

Full Code:

Python

```

import datetime
import random

jokes = [
    "Why do programmers prefer dark mode? Because light attracts bugs!",
    "Python is like English, but for computers!",

```

```

]

def chatbot():
    print("Namaste! Main tumhara simple assistant hoon.")
    while True:
        msg = input("\nTum bolo: ").lower()
        if "hello" in msg or "hi" in msg:
            print("Hello bhai! Kaise ho?")
        elif "time" in msg:
            print("Current time:", datetime.datetime.now().strftime("%H:%M:%S"))
        elif "joke" in msg:
            print(random.choice(jokes))
        elif "add" in msg or "calculate" in msg:
            try:
                expr = input("Calculation likho (jaise 5 + 10): ")
                print("Answer =", eval(expr))
            except:
                print("Galat calculation!")
        elif "bye" in msg or "exit" in msg:
            print("Bye bhai! Phir milte hain!")
            break
        else:
            print("Sorry, samajh nahi aaya!")

chatbot()

```

